

Hybrid Deconvolution Software Handbook

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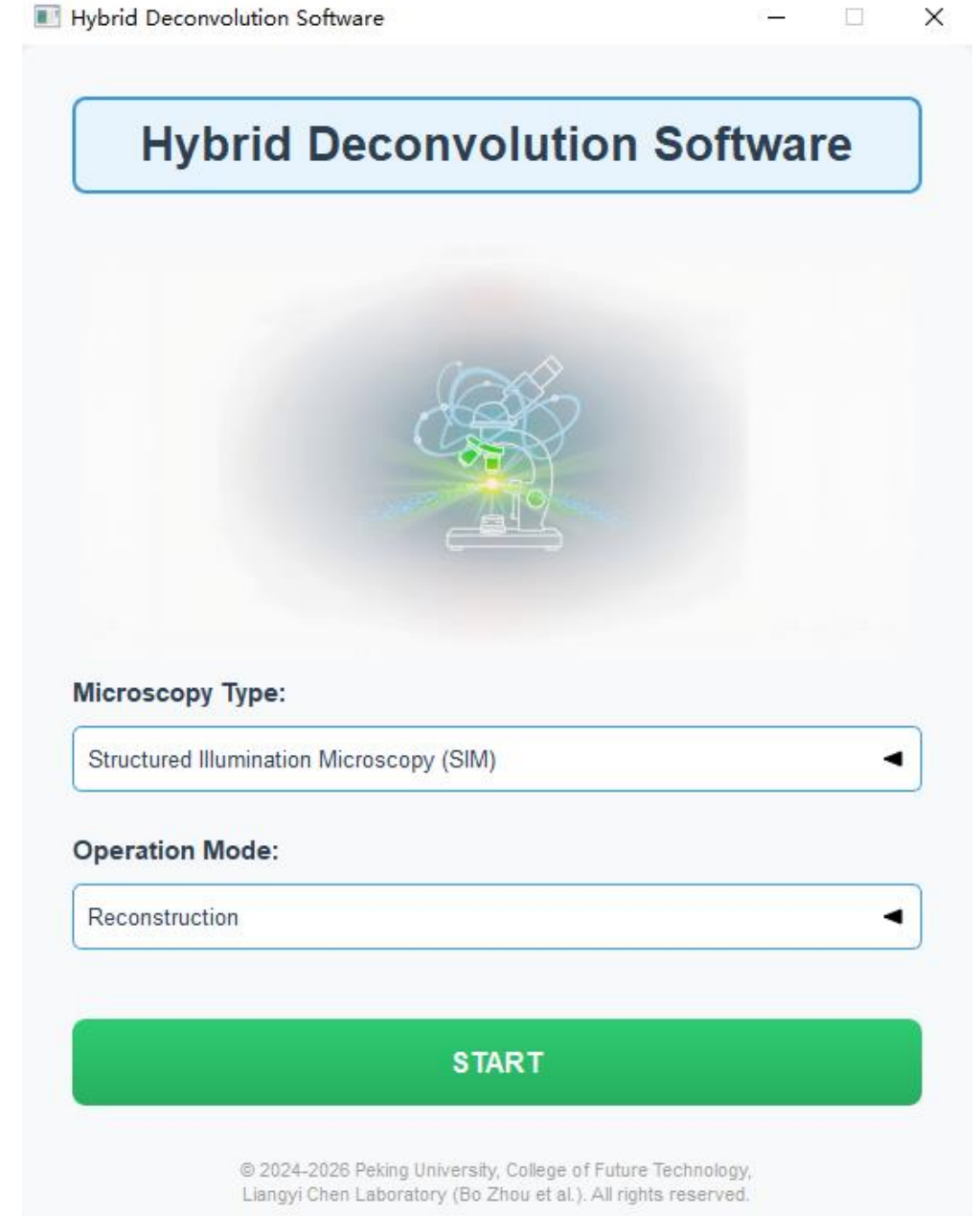
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Hybrid Deconvolution Software: Create Environment and Launch Graphical User Interface (GUI)

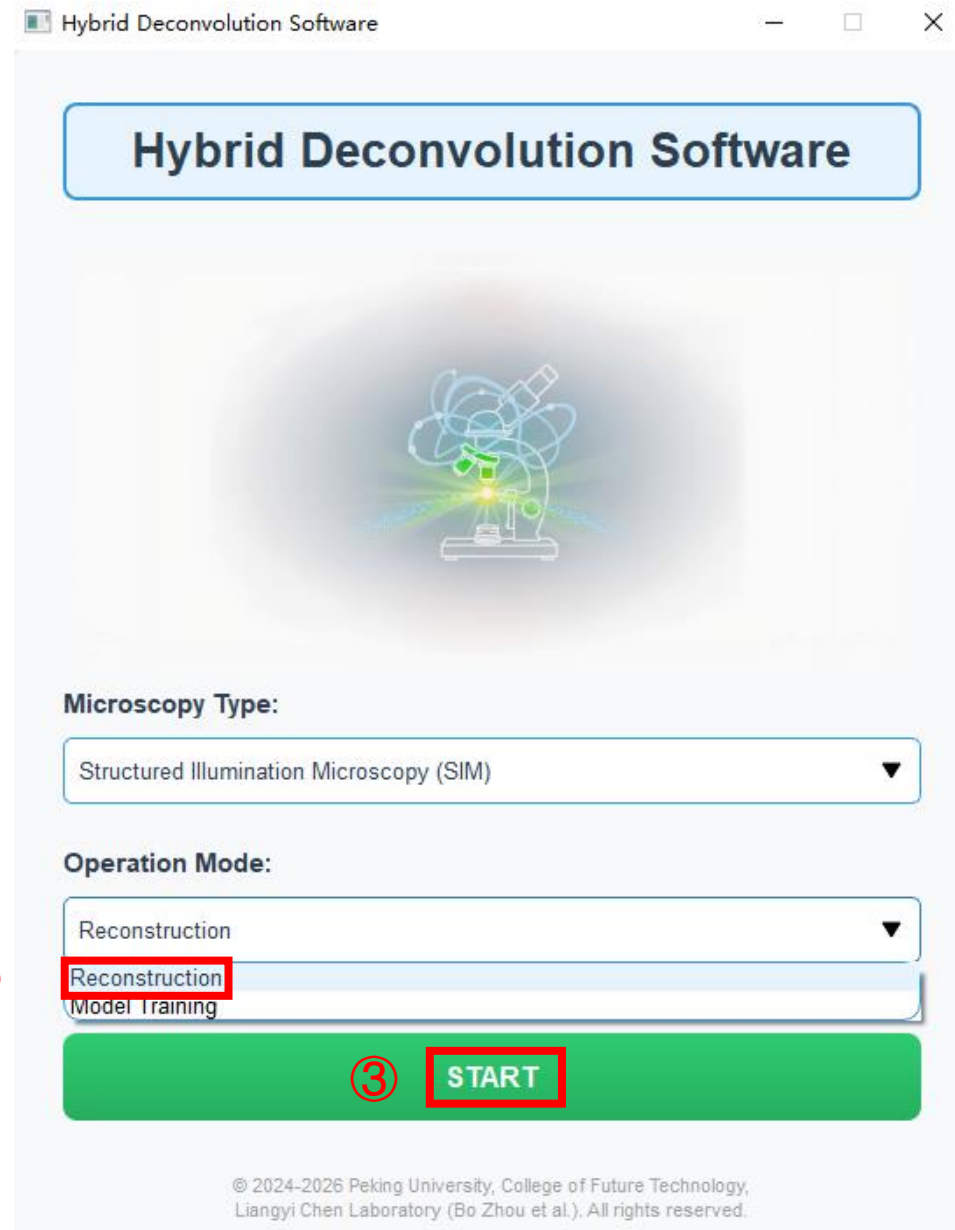
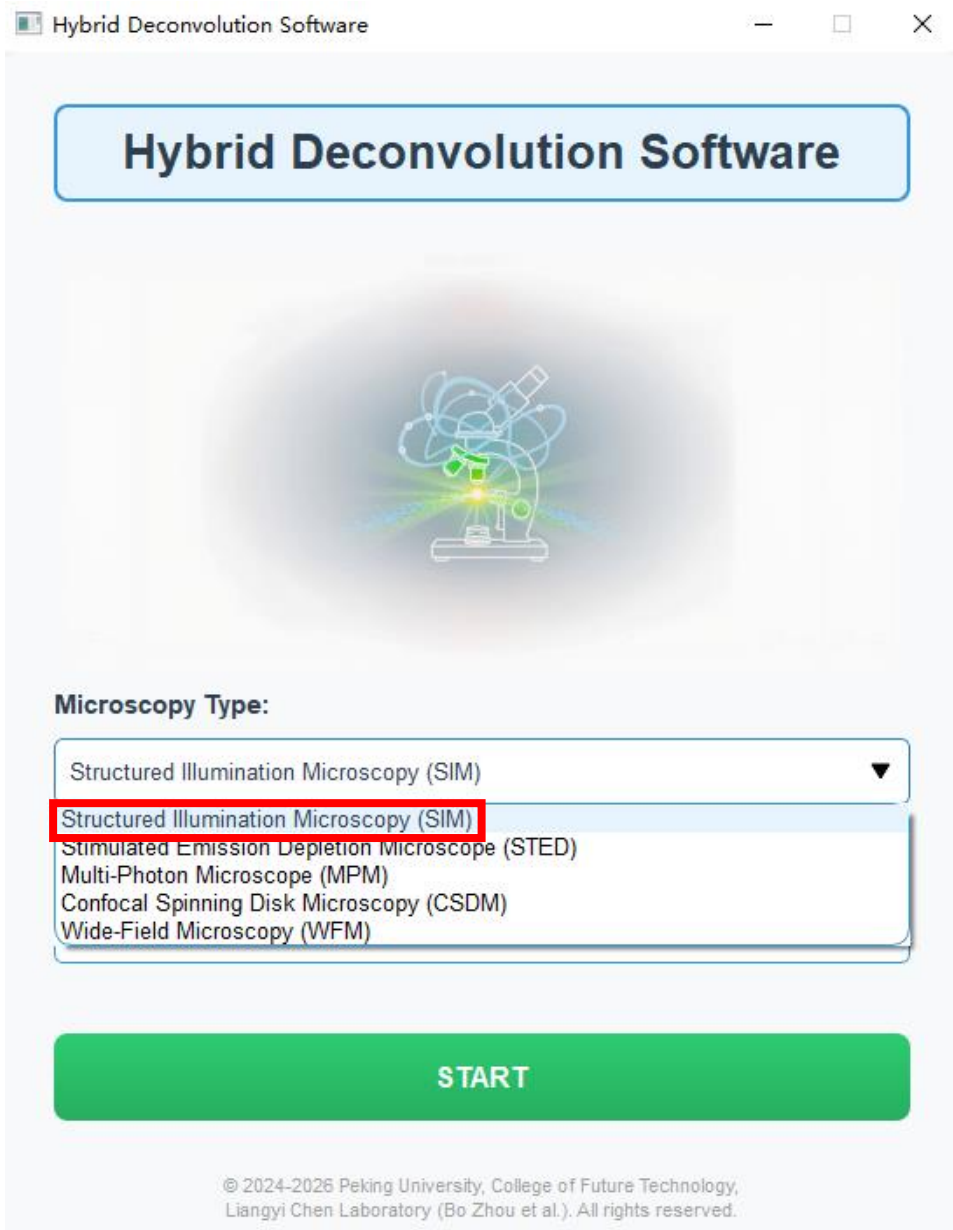
1. Install Anaconda (<https://www.anaconda.com>);
2. Please unzip the Hybrid Deconvolution Software package, **navigate to its root directory in your command prompt**, then sequentially run the following commands to create the Conda environment and launch the GUI:
 - `conda create -n Hybrid_Deconvolution python=3.7 -y`
 - **`conda activate Hybrid_Deconvolution`**
 - `conda install cudatoolkit=11.6`
 - `pip install -r requirements.txt`
 - **`python Launch.py`**
3. For subsequent launches, you only need to run the **2nd** and **5th** commands **from the Software root directory**.

Tip: Please adjust the versions of cudatoolkit, cupy-cuda, torch, torchvision, and torchaudio to match your specific GPU if necessary.



Hybrid-SIM Reconstruction

Start Hybrid-SIM Reconstruction GUI



Hybrid-SIM Reconstruction GUI Layout

① Select parameters

② Run

③ Wait for progress

④ Visualize and export results

The screenshot displays the 'Hybrid-SIM Reconstruction' GUI. The interface is divided into two main panels: 'Hybrid-SIM Reconstruction Parameters' on the left and 'Reconstruction Results' on the right.

Hybrid-SIM Reconstruction Parameters Panel:

- At the top, there are four tabs: 'BF-SIM', 'Hessian Denoise', 'TDV Denoise', and 'Sparse Deconvolution'. The 'BF-SIM' tab is currently selected.
- Below the tabs, the 'BF-SIM Parameters' section contains several input fields and buttons:
 - 'SIM Raw Data Path': A text field with 'no_Recon_SIM.tif' and 'Browse' and 'Reset' buttons.
 - 'SIM Excitation NA': A numeric input field with '1.40' and a 'Reset' button.
 - 'SIM Emission NA': A numeric input field with '1.40' and a 'Reset' button.
 - 'SIM Excitation Wavelength (nm)': A numeric input field with '488.00' and a 'Reset' button.
 - 'SIM Emission Wavelength (nm)': A numeric input field with '525.00' and a 'Reset' button.
 - 'SIM Raw Pixel Size (nm)': A numeric input field with '43.33' and a 'Reset' button.
 - 'SIM Recon OTF Path': A text field with 'Simulation' and 'Browse' and 'Reset' buttons.
- Below the parameters, there is a 'Device:' dropdown menu showing 'cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)' and a 'Detected: 3 device(s)' label.
- At the bottom of the parameters section are three buttons: 'Default Parameters', 'Load Parameters', and 'Export Parameters'.
- Below the parameters section is a large green button labeled 'Run Reconstruction'.
- Below the 'Run Reconstruction' button is a grey button labeled 'Stop Reconstruction'.
- At the bottom of the panel is a green progress bar showing '100%' and the text 'Reconstruction completed!'.

Reconstruction Results Panel:

- At the top, there are two tabs: 'BF-SIM Reconstruction' and 'BF+Hessian-SIM Reconstruction'. The 'BF-SIM Reconstruction' tab is currently selected.
- Below the tabs, there are four image displays showing reconstruction results:
 - 'BF-SIM Reconstruction': A grayscale image showing a dense network of fibers.
 - 'BF+Hessian-SIM Reconstruction': A grayscale image showing a dense network of fibers.
 - 'BF+Hessian+TDV-SIM Reconstruction': A grayscale image showing a dense network of fibers.
 - 'Hybrid-SIM Reconstruction': A grayscale image showing a dense network of fibers.
- Below the image displays, there is a 'Display Range' slider and a 'Frame' slider.
- At the bottom right of the results panel are three buttons: 'Reset Range', 'Reset View', and 'Export Results'.

Hybrid-SIM Reconstruction Parameters ---- BF-SIM Parameters

Hybrid-SIM Reconstruction Parameters

BF-SIM

Hessian Denoise

TDV Denoise

Sparse Deconvolution

BF-SIM Parameters

SIM Raw Data Path	no_Recon_SIM.tif	Browse	Reset
SIM Excitation NA	1.40	Reset	
SIM Emission NA	1.40	Reset	
SIM Excitation Wavelength (nm)	488.00	Reset	
SIM Emission Wavelength (nm)	525.00	Reset	
SIM Raw Pixel Size (nm)	43.33	Reset	
SIM Recon OTF Path	Simulation	Browse	Reset

#: Please prepare the raw data TIFF file in advance.

SIM Raw Data Path:

1. TIFF file path of the SIM raw data#;
2. Please choose the file via the Browse button;
3. Sample file: [Demo_Recon_SIM.tif].

SIM Excitation/Emission NA:

1. Excitation/Emission numerical aperture (NA) of the SIM optical system;
2. Please set the values slightly lower than the NA labeled on the objective lens.

SIM Excitation/Emission Wavelength:

1. Excitation/Emission wavelength during the SIM imaging;
2. Please set the values based on the properties of the fluorophore.

SIM Raw Pixel Size:

1. Pixel size of the SIM raw data;
2. Please set the value to the camera pixel's physical size divided by the SIM magnification ratio.

SIM Recon OTF Path:

1. TIFF file path of SIM optical transfer function path;
2. Please select an experimentally measured OTF TIFF file via the Browse button to replace the current default simulated OTF;
3. Sample experimental OTF files are in [src_Hybrid\src_Optics] subfolder within the software installation directory.

Hybrid-SIM Reconstruction Parameters ---- Hessian Denoise Parameters

Hybrid-SIM Reconstruction Parameters

BF-SIM

Hessian Denoise

TDV Denoise

Sparse Deconvolution

Hessian Denoise Parameters

Hessian Fidelity

100.00

Reset

Hessian Z Continuity

2.00

Reset

Hessian Iteration Number

100

Reset

Hessian Fidelity:

1. Fidelity term weight of the Hessian optimization function;
2. Please set the value lower for lower signal-to-noise ratio (SNR) data.

Hessian Z Continuity:

1. The weight for temporal/axial continuity in the Hessian regularization term;
2. Please set the value higher for well-sampled time-lapse or volumetric data.

Hessian Iteration Number:

1. Number of iterations for solving the problem constrained by the Hessian regularization term;
2. Please set the value higher for lower SNR data;
3. Please set the value to 0 for skipping the Hessian Denoise.

Tip: For lower SNR data, please firstly adjust the [Hessian Fidelity], then adjust the [Hessian Iteration Number] if necessary.

Hybrid-SIM Reconstruction Parameters ---- TDV Denoise Parameters

Hybrid-SIM Reconstruction Parameters

BF-SIM

Hessian Denoise

TDV Denoise

Sparse Deconvolution

TDV Denoise Parameters

TDV Model Path

50XSIM_Actin.pth

Browse

Reset

TDV Fidelity

100.00



Reset

TDV Offset

0.10



Reset

TDV Weight

1.00



Reset

TDV Iteration Number

30



Reset

TDV Model Path:

1. PTH file path of the TDV-DNN model;
2. Please choose the file via the Browse button;
3. Please train a new model using the [**Model Training**] mode.

TDV Fidelity:

1. Fidelity term weight of the TDV optimization function;
2. Please set the value lower for lower SNR data.

TDV Offset:

1. Offset value for normalizing the input data;
2. Please set the value higher if apparent signal loss is observed.

TDV Weight:

1. TDV regularization term weight of the TDV optimization function;
2. Please set the value higher for lower SNR data.

TDV Iteration Number:

1. Number of iterations for solving the problem constrained by the TDV regularization term;
2. Please set the value higher for lower SNR data;
3. Please set the value to 0 for skipping the TDV Denoise.

Tip: For low SNR data, please firstly adjust the [TDV Fidelity], then adjust the [TDV Weight] and [TDV Iteration Number] if necessary.

Hybrid-SIM Reconstruction Parameters ---- Sparse Deconvolution Parameters

Hybrid-SIM Reconstruction Parameters

BF-SIM

Hessian Denoise

TDV Denoise

Sparse Deconvolution

Sparse Deconvolution Parameters

Sparse NA

3.00

Reset

Sparse Offset

0.20

Reset

Sparse Iteration Number

25

Reset

Sparse NA:

1. The equivalent NA assumed during point spread function (PSF) generation for sparse deconvolution.
2. Please set the value to ~ 0.8 –2 times the emission NA.

TDV Offset:

1. Offset value for normalizing the input data;
2. Please set the value higher if apparent signal loss is observed.

Sparse Iteration Number:

1. Number of iterations for solving the deconvolution problem constrained by the Sparsity regularization term;
2. Please set the value higher for better resolution improvement;
3. Please set the value to 0 for skipping the Sparse Deconvolution.

Tip: Please begin by setting the [Sparse Iteration Number] (~ 10 –20) and [TDV Offset] (~ 0.1 –0.2). Next, determine the optimal [Sparse NA]. Finally, fine-tune the first two parameters to balance resolution improvement against signal preservation.

Hybrid-SIM Reconstruction Parameters ---- Device

Please select the computing device name (cpu/cuda) for reconstruction.

Hybrid-SIM Reconstruction

Hybrid-SIM Reconstruction Parameters

BF-SIM

Hessian Denoise

TDV Denoise

Sparse Deconvolution

BF-SIM Parameters

SIM Raw Data Path

no_Recon_SIM.tif

Browse

Reset

SIM Excitation NA

1.40

Reset

SIM Emission NA

1.40

Reset

SIM Excitation Wavelength (nm)

488.00

Reset

SIM Emission Wavelength (nm)

525.00

Reset

SIM Raw Pixel Size (nm)

43.33

Reset

SIM Recon OTF Path

Simulation

Browse

Reset

Device:

cpu

cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)

cuda:1 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)

Detected: 3 device(s)

Default Parameters

Load Parameters

Export Parameters

Run Reconstruction

Stop Reconstruction

100%

Reconstruction completed !

Reconstruction Results

BF-SIM Reconstruction

BF+Hessian-SIM Reconstruction

BF+Hessian+TDV-SIM Reconstruction

Hybrid-SIM Reconstruction

Display Range:

Reset Range

Reset View

Export Results

Frame:

7/10

Hybrid-SIM Reconstruction Parameters ---- Default, Load and Export Parameters

Please click the button to export the current parameters.

Please click the button to reset the default parameters.

Please click the button to load the previously exported parameters.

The screenshot displays the 'Hybrid-SIM Reconstruction' software interface. The main window is titled 'Hybrid-SIM Reconstruction Parameters' and features four tabs: 'BF-SIM', 'Hessian Denoise', 'TDV Denoise', and 'Sparse Deconvolution'. The 'BF-SIM' tab is active, showing parameters for 'BF-SIM Parameters'. These parameters include 'SIM Raw Data Path' (no_Recon_SIM.tif), 'SIM Excitation NA' (1.40), 'SIM Emission NA' (1.40), 'SIM Excitation Wavelength (nm)' (488.00), 'SIM Emission Wavelength (nm)' (525.00), 'SIM Raw Pixel Size (nm)' (43.33), and 'SIM Recon OTF Path' (Simulation). Each parameter has a 'Reset' button. Below the parameters, there is a 'Device' dropdown menu set to 'cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)' and a 'Detected: 3 device(s)' label. At the bottom of the parameter section, there are three buttons: 'Default Parameters', 'Load Parameters', and 'Export Parameters'. Below these are two large buttons: 'Run Reconstruction' (green) and 'Stop Reconstruction' (grey). A green progress bar at the bottom indicates '100%' completion, with the text 'Reconstruction completed!' below it. To the right of the parameter section is the 'Reconstruction Results' panel, which displays four images: 'BF-SIM Reconstruction', 'BF+Hessian-SIM Reconstruction', 'BF+Hessian+TDV-SIM Reconstruction', and 'Hybrid-SIM Reconstruction'. At the bottom of the results panel, there are sliders for 'Display Range' and 'Frame' (set to 7/10), and buttons for 'Reset Range', 'Reset View', and 'Export Results'.

Hybrid-SIM Reconstruction Parameters

BF-SIM Hessian Denoise TDV Denoise Sparse Deconvolution

BF-SIM Parameters

SIM Raw Data Path no_Recon_SIM.tif Browse Reset

SIM Excitation NA 1.40 Reset

SIM Emission NA 1.40 Reset

SIM Excitation Wavelength (nm) 488.00 Reset

SIM Emission Wavelength (nm) 525.00 Reset

SIM Raw Pixel Size (nm) 43.33 Reset

SIM Recon OTF Path Simulation Browse Reset

Device: cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB) Detected: 3 device(s)

Default Parameters Load Parameters Export Parameters

Run Reconstruction

Stop Reconstruction

100%

Reconstruction completed!

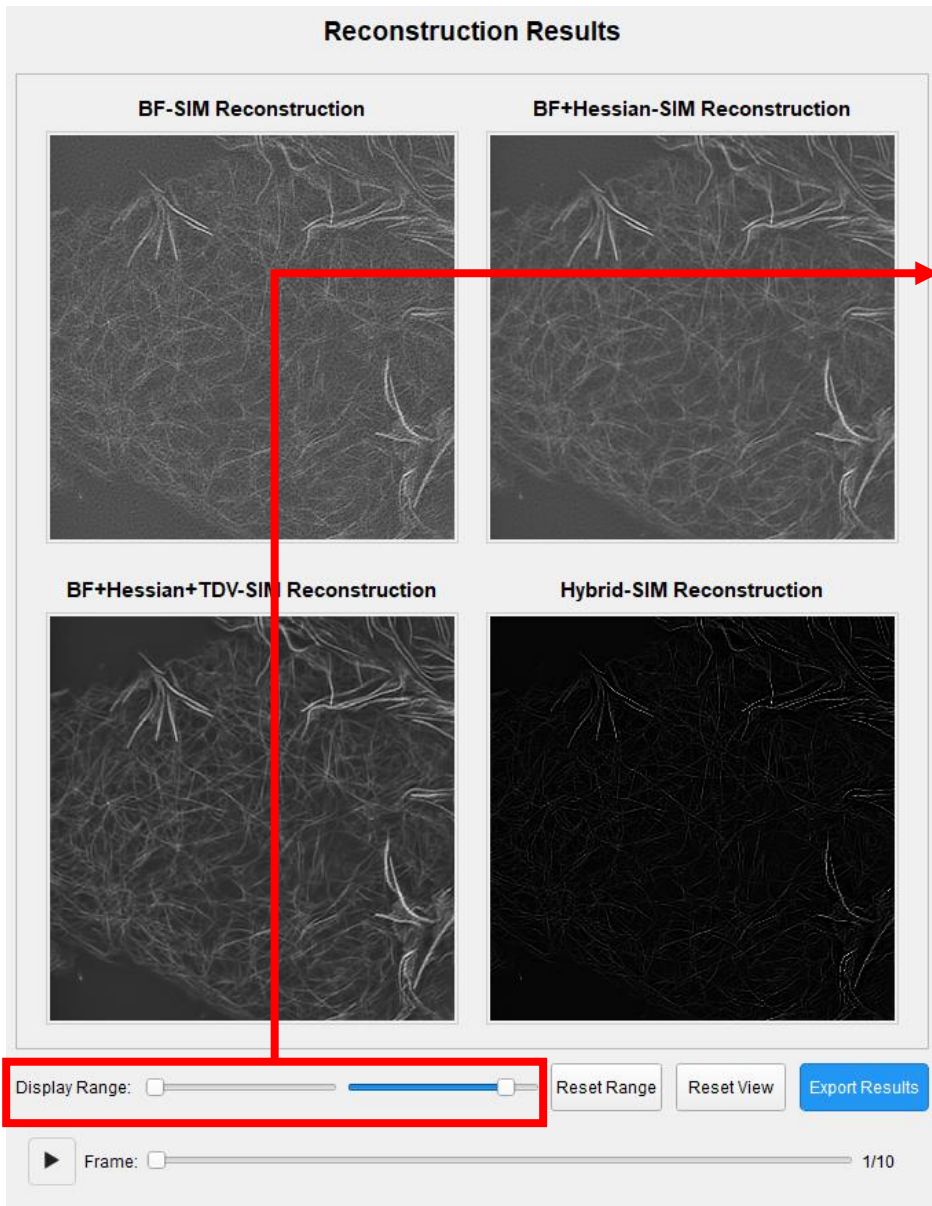
Reconstruction Results

BF-SIM Reconstruction BF+Hessian-SIM Reconstruction

BF+Hessian+TDV-SIM Reconstruction Hybrid-SIM Reconstruction

Display Range: Reset Range Reset View Export Results

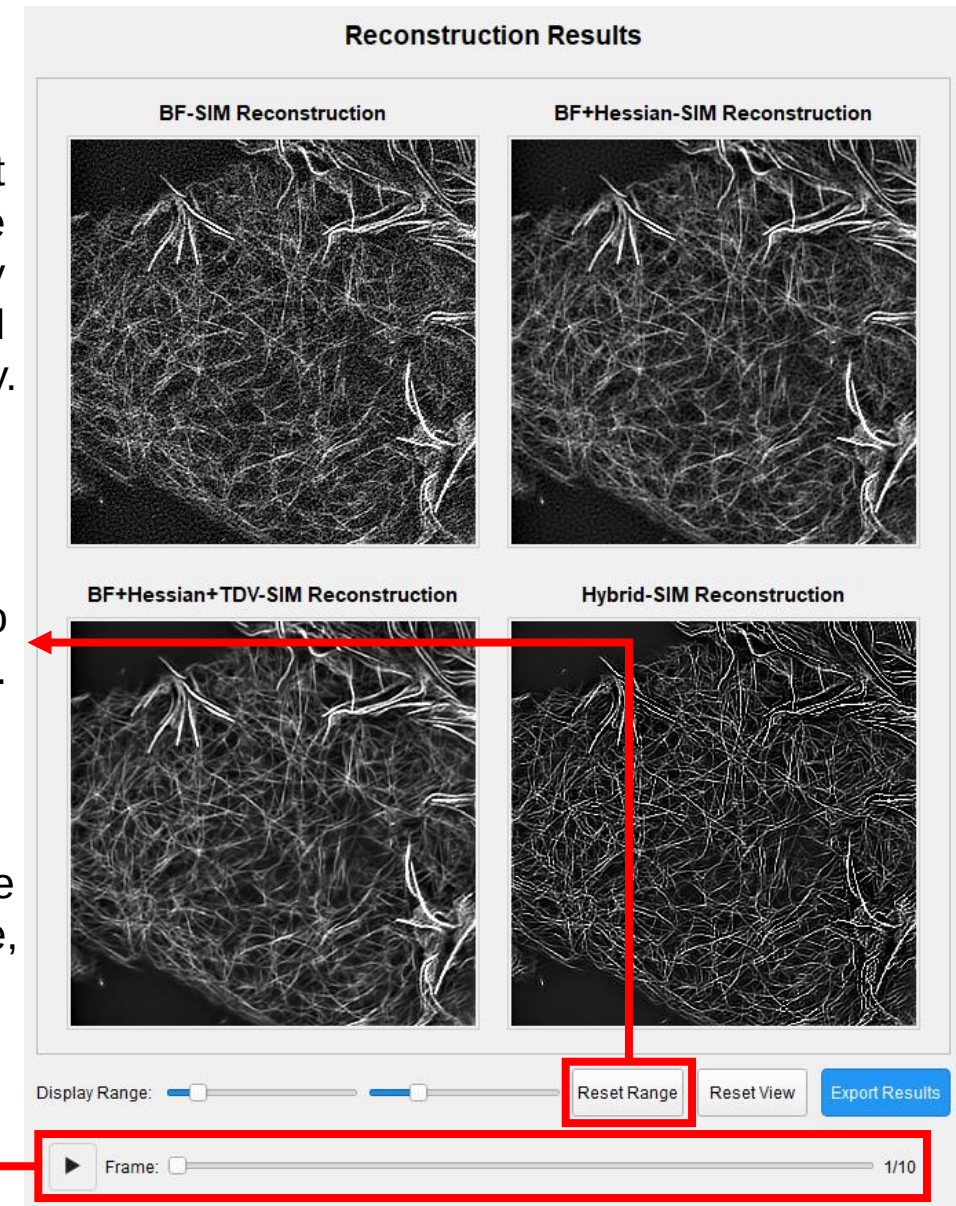
Frame: 7/10



Please use the left and right sliders here to adjust the image display range by setting the minimum and maximum values, respectively.

Please click the button to
reset the image display range.

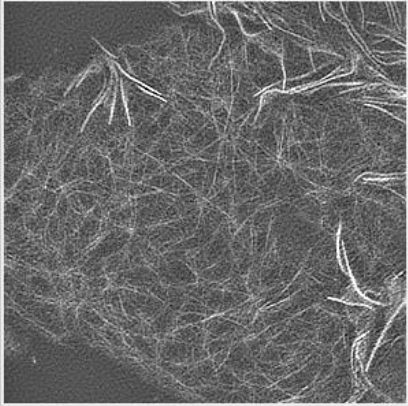
Please use the slider to browse reconstruction results by frame, and click the left button to play the video.



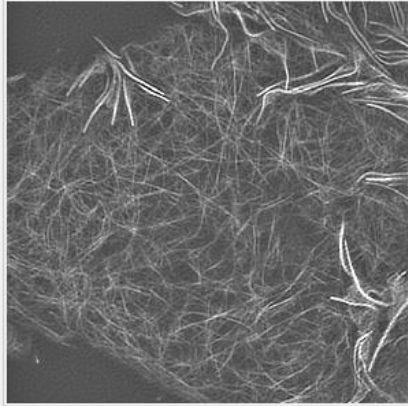
Hybrid-SIM Reconstruction Results Visualization ---- Zoom View

Reconstruction Results

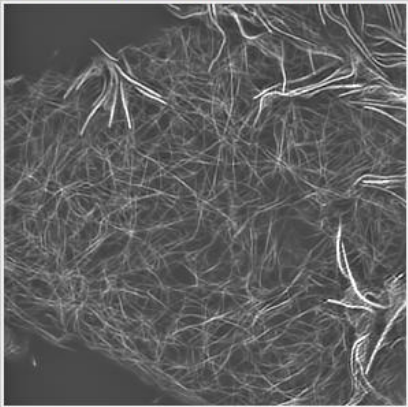
BF-SIM Reconstruction



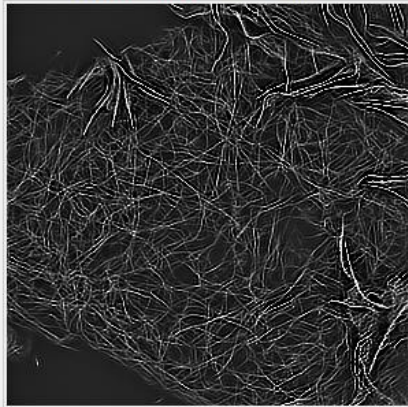
BF+Hessian-SIM Reconstruction



BF+Hessian+TDV-SIM Reconstruction



Hybrid-SIM Reconstruction



Display Range:

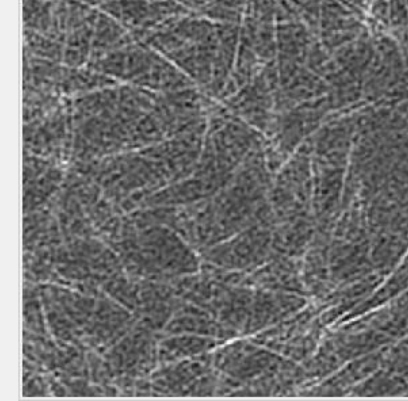
Frame: 3/10

Please scroll the mouse wheel within any area of these four images to zoom the view.

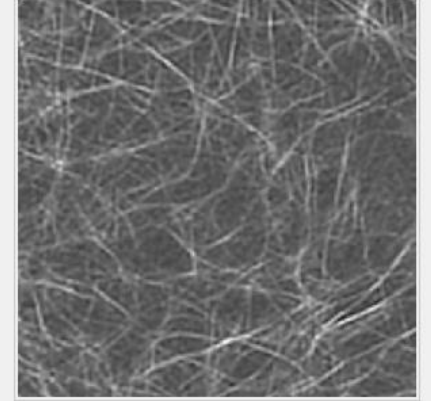
Please click the button to reset the image zoom view.

Reconstruction Results

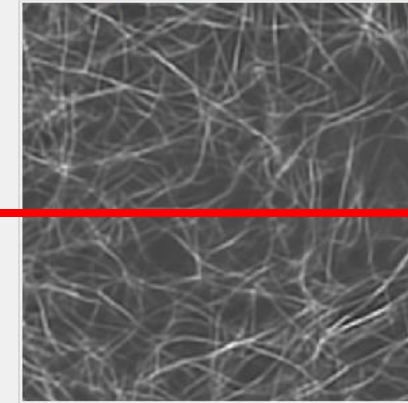
BF-SIM Reconstruction



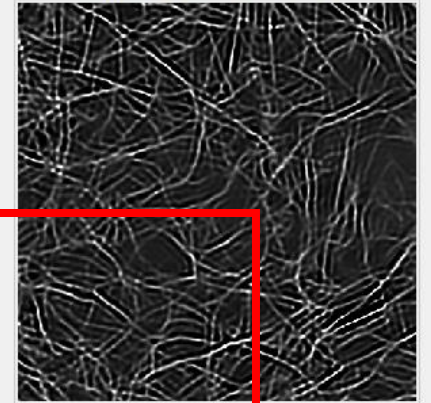
BF+Hessian-SIM Reconstruction



BF+Hessian+TDV-SIM Reconstruction



Hybrid-SIM Reconstruction



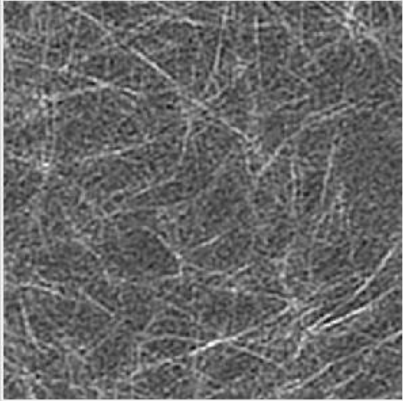
Display Range:

Frame: 3/10

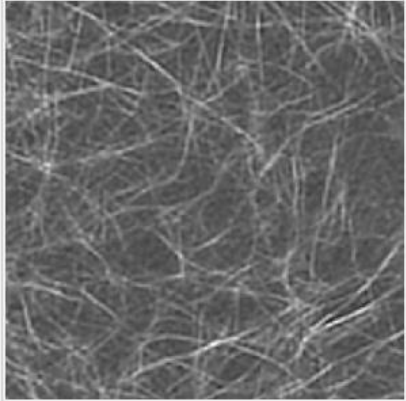
Hybrid-SIM Reconstruction Results Export

Reconstruction Results

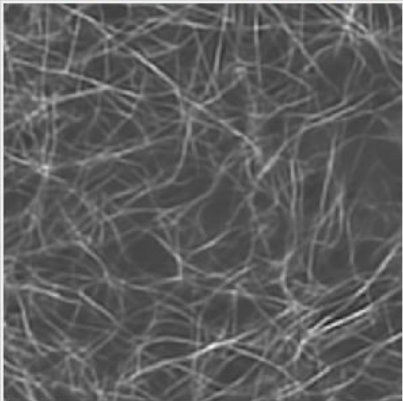
BF-SIM Reconstruction



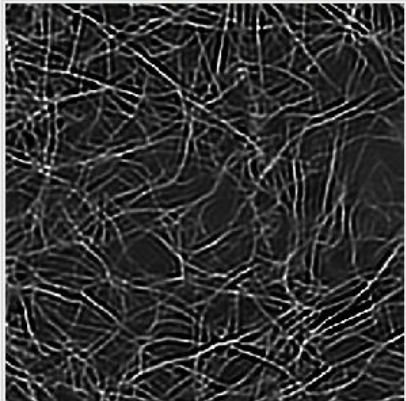
BF+Hessian-SIM Reconstruction



BF+Hessian+TDV-SIM Reconstruction



Hybrid-SIM Reconstruction



Display Range:  Reset Range Reset View **Export Results**

Frame:  3/10

Export Results

Export Full Reconstruction

- ☐ BF-SIM Reconstruction
- ☐ BF+Hessian-SIM Reconstruction
- ☐ BF+Hessian+TDV-SIM Reconstruction
- ☒ Hybrid-SIM Reconstruction

Export Single-Frame Comparison

- ☐ Export Comparison

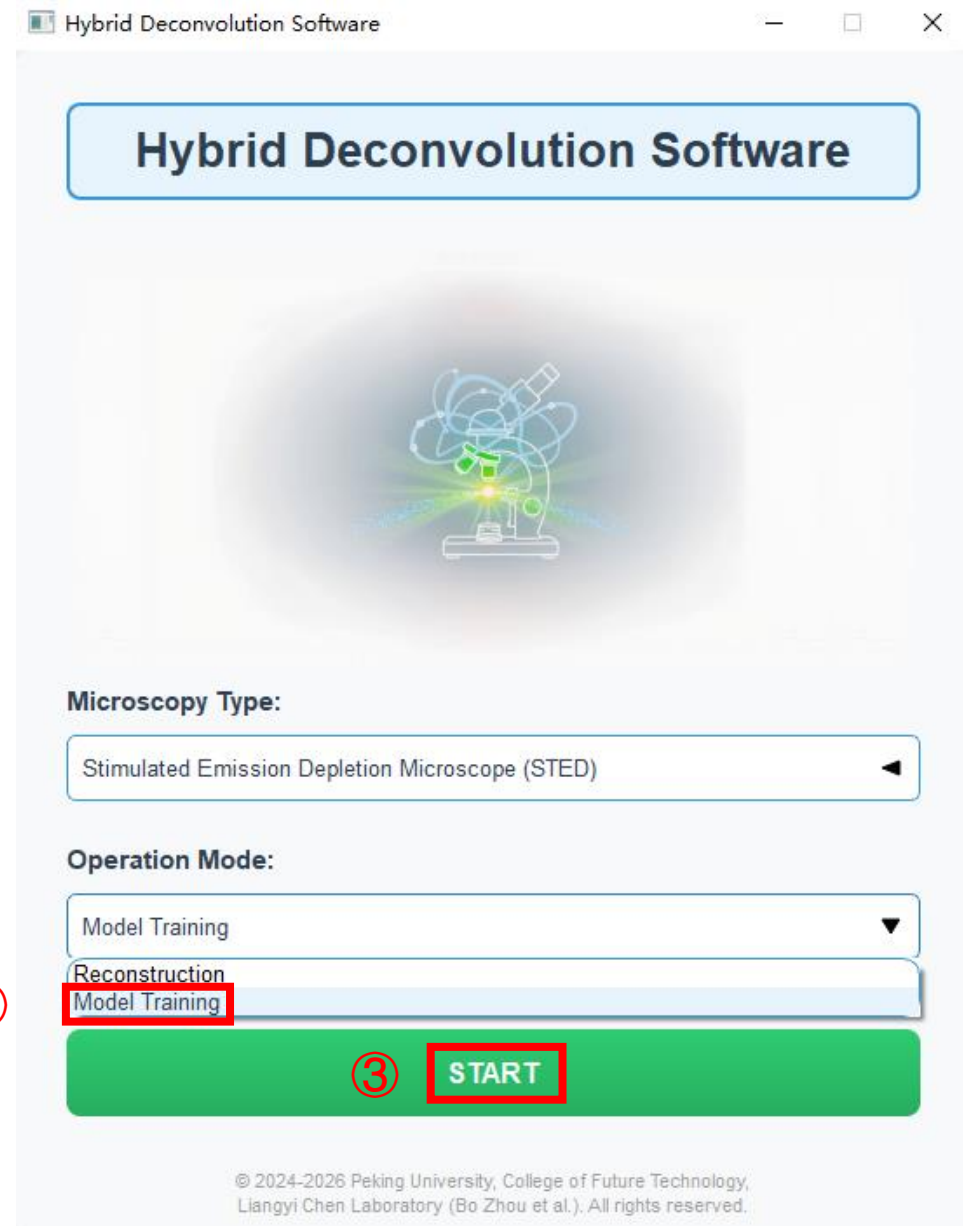
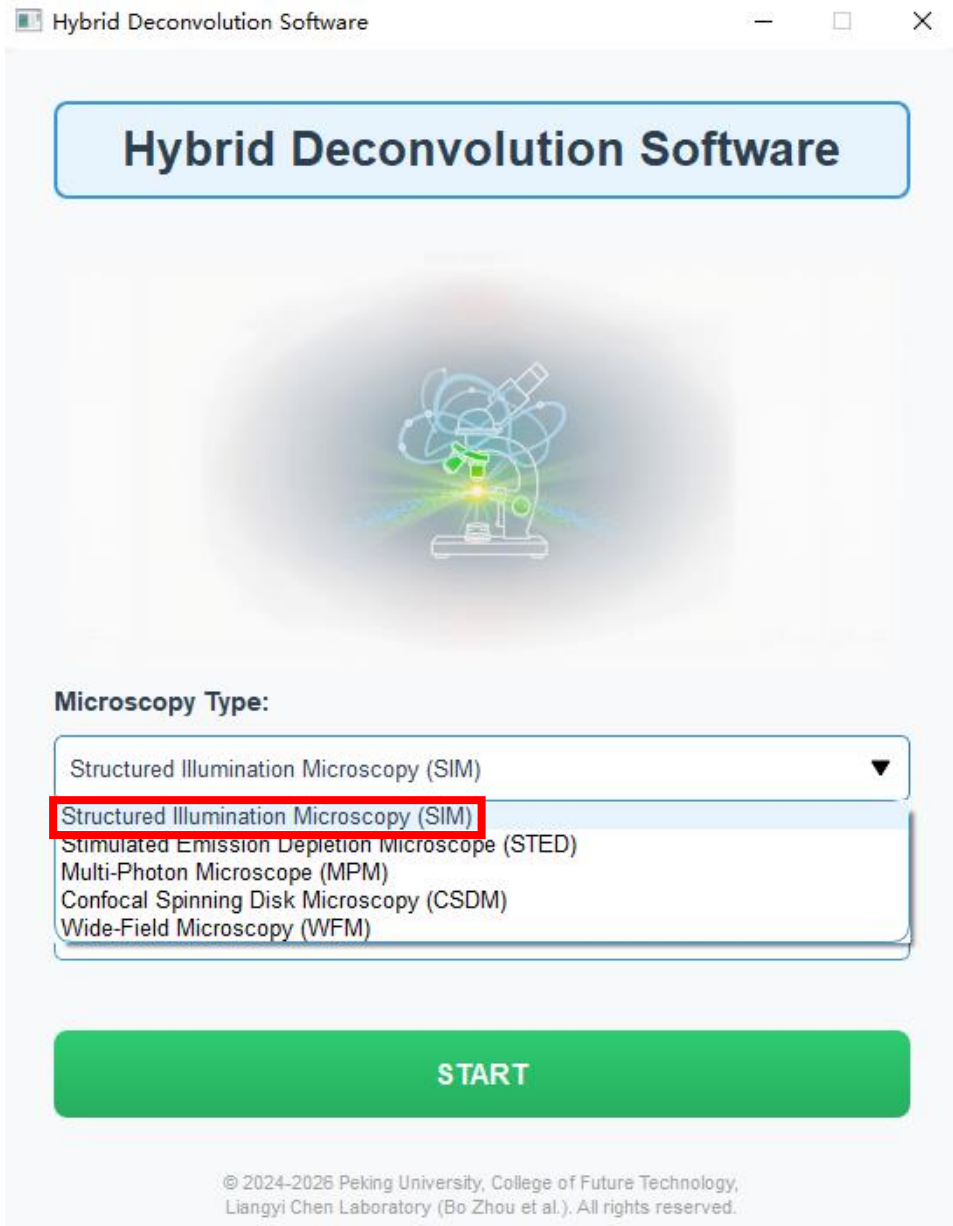
Frame index: 1 Range: 1 - 10

Note: The comparison will sequentially contain BF, BF+Hessian, BF+Hessian+TDV and Hybrid reconstructions.

OK Cancel

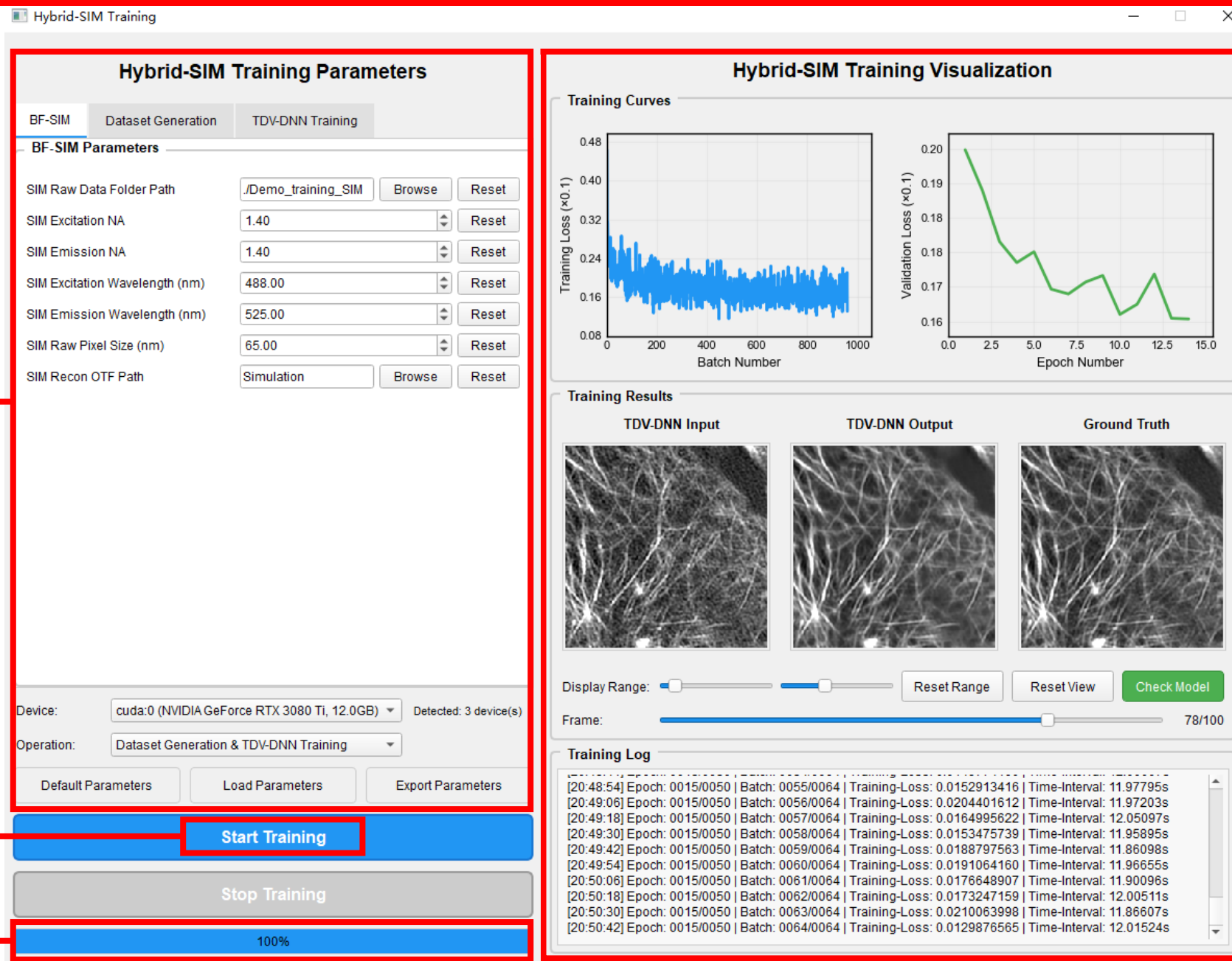
Hybrid-SIM Model Training

Start Hybrid-SIM Model Training GUI



Hybrid-SIM Training GUI layout

① Select parameters



② Run

③ Wait for progress

④ Visualize training process and export TDV-DNN model

Hybrid-SIM Training Parameters ---- BF-SIM Parameters

Hybrid-SIM Training Parameters

BF-SIM

Dataset Generation

TDV-DNN Training

BF-SIM Parameters

SIM Raw Data Folder Path	/Demo_training_SIM	Browse	Reset
SIM Excitation NA	1.40	Reset	
SIM Emission NA	1.40	Reset	
SIM Excitation Wavelength (nm)	488.00	Reset	
SIM Emission Wavelength (nm)	525.00	Reset	
SIM Raw Pixel Size (nm)	65.00	Reset	
SIM Recon OTF Path	Simulation	Browse	Reset

#: Please prepare the raw data folder in advance.

SIM Raw Data Folder Path:

1. Folder path of the SIM raw data TIFF files#;
2. Please choose the folder via the Browse button;
3. Sample folder: [Demo_training_SIM].

SIM Excitation/Emission NA:

1. Excitation/Emission numerical aperture (NA) of the SIM optical system;
2. Please set the values slightly lower than the NA labeled on the objective lens.

SIM Excitation/Emission Wavelength:

1. Excitation/Emission wavelength during the SIM imaging;
2. Please set the values based on the properties of the fluorophore.

SIM Raw Pixel Size:

1. Pixel size of the SIM raw data;
2. Please set the value to the camera pixel's physical size divided by the SIM magnification ratio.

SIM Recon OTF Path:

1. TIFF file path of SIM optical transfer function path;
2. Please select an experimentally measured OTF TIFF file via the Browse button to replace the current default simulated OTF;
3. Sample experimental OTF files are in [src_Hybrid\src_Optics] subfolder within the software installation directory.

Hybrid-SIM Training Parameters ---- Dataset Generation Parameters

Hybrid-SIM Training Parameters

BF-SIM

Dataset Generation

TDV-DNN Training

Dataset Generation Parameters

Dataset Z Average Number (frame)	20	Reset
Dataset Z Skip Number (frame)	10	Reset
Dataset XY Block Size (pixel)	256	Reset
Dataset XY Block Interval (pixel)	128	Reset
Dataset XY Poisson Noise Level	50.00	Reset
Dataset XY Gaussian Noise Level	50.00	Reset
Dataset XY Minimal Heterogeneity	0.60	Reset

Tip: The generated dataset will be saved in a new folder that appends [_Dataset] to [**SIM Raw Data Folder Path**] (e.g., [Demo_training_SIM_Dataset]).

Dataset Z Average/Skip Number:

1. Average/Skip super-resolution (SR) frame number along the temporal/axial direction of the time-lapse or volumetric data during dataset generation;
2. Please set the [Average Number] higher for generating higher SNR ground-truth (GT) images;
3. Please set the [Skip Number] higher to ensure the dataset contains a sufficient number of images.

Dataset XY Block Size/Interval:

1. Block size and interval for lateral cropping during dataset generation;
2. Please set the [Block Interval] lower to ensure the dataset contains a sufficient number of images.

Dataset XY Poisson/Gaussian Noise Level:

1. Level of Poisson/Gaussian noise added to the averaged SIM raw data during dataset generation;
2. Please set the values higher to improve model denoising capacity;
3. Please set the values lower to avoid model overfitting.

Dataset XY Minimal Heterogeneity:

1. The heterogeneity of all images in the dataset need to exceed this threshold to ensure sufficient sample information;
2. Please set the value higher for dense samples (like actin filaments) and lower for sparse samples (like peroxisomes).

Hybrid-SIM Training Parameters ---- TDV-DNN Training Parameters

Hybrid-SIM Training Parameters

BF-SIM

Dataset Generation

TDV-DNN Training

TDV-DNN Training Parameters

Training System

100XSIM

Reset

Training Sample

Actin

Reset

Training Batch Size

10

Reset

Training Epoch Number

50

Reset

Training Learning Rate

5.00

Reset

Training System/Sample:

1. Imaging system/sample name of the raw data;
2. The trained model will be saved with a filename that includes the system name, sample name, and training timestamp. Such as the [TDV_100XSIM_Actin_20260116_092521.pth], where [100XSIM] is the system, [Actin] is the sample, [20260116_092521] represents the training start time (YYYYMMDD_HHMMSS), and [.pth] is the file format.

Training Batch Size/Epoch Number/Learning Rate:

1. Batch size, epoch number and initial learning rate of the TDV-DNN training process;
2. Please configure the [Batch Size] to maximize video random access memory usage (VRAM) without overflow;
3. Please configure the [Learning Rate] to balance training efficiency and stability.

Hybrid-SIM Training Parameters ---- Device

Please select the computing device name (cpu/cuda) for training.

Hybrid-SIM Training

Hybrid-SIM Training Parameters

BF-SIM

Dataset Generation

TDV-DNN Training

BF-SIM Parameters

SIM Raw Data Folder Path

./Demo_training_SIM

Browse

Reset

SIM Excitation NA

1.40

Reset

SIM Emission NA

1.40

Reset

SIM Excitation Wavelength (nm)

488.00

Reset

SIM Emission Wavelength (nm)

525.00

Reset

SIM Raw Pixel Size (nm)

65.00

Reset

SIM Recon OTF Path

Simulation

Browse

Reset

Device:

cpu

cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)

cuda:1 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)

Detected: 3 device(s)

Default Parameters

Load Parameters

Export Parameters

Start Training

Stop Training

100%

Hybrid-SIM Training Visualization

Training Curves

Training Loss ($\times 0.1$)

Batch Number

Validation Loss ($\times 0.1$)

Epoch Number

Training Results

TDV-DNN Input

TDV-DNN Output

Ground Truth

Display Range:

Reset Range

Reset View

Check Model

Frame:

78/100

Training Log

[20:48:54] Epoch: 0015/0050 | Batch: 0055/0064 | Training-Loss: 0.0152913416 | Time-Interval: 11.97795s

[20:49:06] Epoch: 0015/0050 | Batch: 0056/0064 | Training-Loss: 0.0204401612 | Time-Interval: 11.97203s

[20:49:18] Epoch: 0015/0050 | Batch: 0057/0064 | Training-Loss: 0.0164995622 | Time-Interval: 12.05097s

[20:49:30] Epoch: 0015/0050 | Batch: 0058/0064 | Training-Loss: 0.0153475739 | Time-Interval: 11.95895s

[20:49:42] Epoch: 0015/0050 | Batch: 0059/0064 | Training-Loss: 0.0188797563 | Time-Interval: 11.86098s

[20:49:54] Epoch: 0015/0050 | Batch: 0060/0064 | Training-Loss: 0.0191064160 | Time-Interval: 11.96655s

[20:50:06] Epoch: 0015/0050 | Batch: 0061/0064 | Training-Loss: 0.0176648907 | Time-Interval: 11.90096s

[20:50:18] Epoch: 0015/0050 | Batch: 0062/0064 | Training-Loss: 0.0173247159 | Time-Interval: 12.00511s

[20:50:30] Epoch: 0015/0050 | Batch: 0063/0064 | Training-Loss: 0.0210063998 | Time-Interval: 11.86607s

[20:50:42] Epoch: 0015/0050 | Batch: 0064/0064 | Training-Loss: 0.0129876565 | Time-Interval: 12.01524s

Hybrid-SIM Reconstruction Parameters ---- Default, Load and Export Parameters

Please click the button to export the current parameters.

Please click the button to reset the default parameters.

Please click the button to load the previously exported parameters.

Hybrid-SIM Training Parameters

BF-SIM

Dataset Generation

TDV-DNN Training

BF-SIM Parameters

SIM Raw Data Folder Path

./Demo_training_SIM

Browse

Reset

SIM Excitation NA

1.40

Reset

SIM Emission NA

1.40

Reset

SIM Excitation Wavelength (nm)

488.00

Reset

SIM Emission Wavelength (nm)

525.00

Reset

SIM Raw Pixel Size (nm)

65.00

Reset

SIM Recon OTF Path

Simulation

Browse

Reset

Device:

cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)

Detected: 3 device(s)

Operation:

Dataset Generation & TDV-DNN Training

Default Parameters

Load Parameters

Export Parameters

Start Training

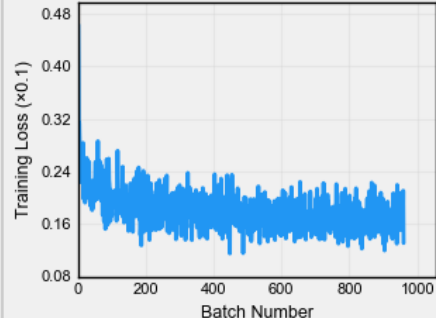
Stop Training

100%

Hybrid-SIM Training Visualization

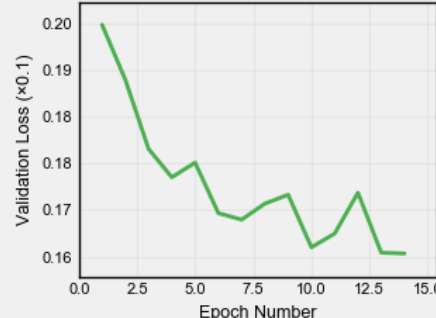
Training Curves

Training Loss ($\times 0.1$)



Batch Number

Validation Loss ($\times 0.1$)



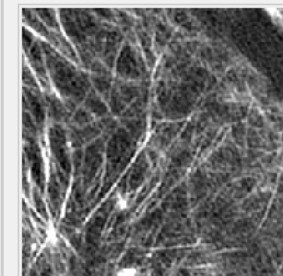
Epoch Number

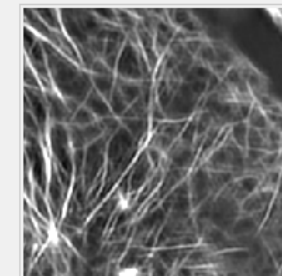
Training Results

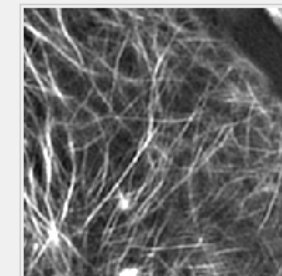
TDV-DNN Input

TDV-DNN Output

Ground Truth







Display Range:

Reset Range

Reset View

Check Model

Frame:

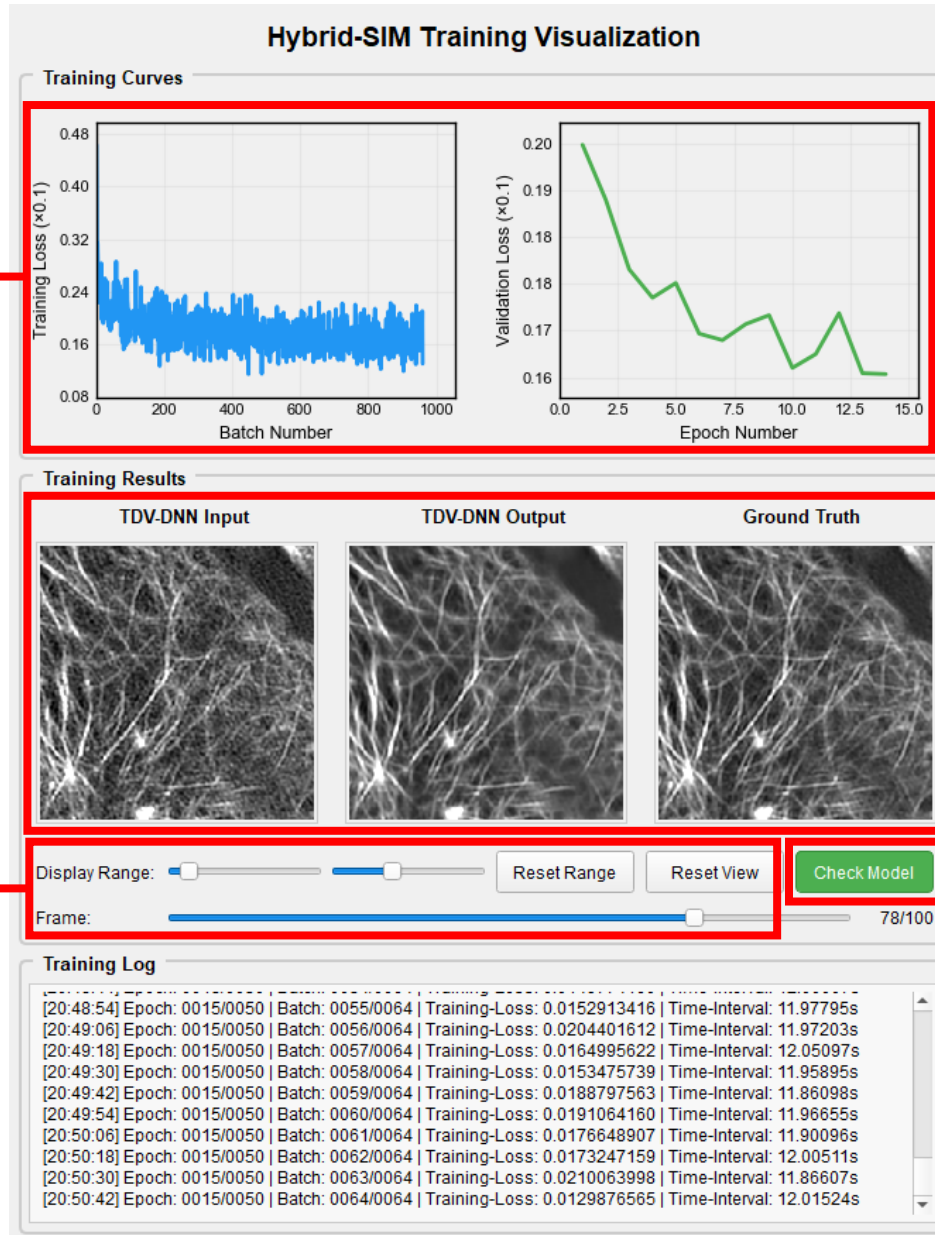
78/100

Training Log

[20:48:54]	Epoch: 0015/0050	Batch: 0055/0064	Training-Loss: 0.0152913416	Time-Interval: 11.97795s
[20:49:06]	Epoch: 0015/0050	Batch: 0056/0064	Training-Loss: 0.0204401612	Time-Interval: 11.97203s
[20:49:18]	Epoch: 0015/0050	Batch: 0057/0064	Training-Loss: 0.0164995622	Time-Interval: 12.05097s
[20:49:30]	Epoch: 0015/0050	Batch: 0058/0064	Training-Loss: 0.0153475739	Time-Interval: 11.95895s
[20:49:42]	Epoch: 0015/0050	Batch: 0059/0064	Training-Loss: 0.0188797563	Time-Interval: 11.86098s
[20:49:54]	Epoch: 0015/0050	Batch: 0060/0064	Training-Loss: 0.0191064160	Time-Interval: 11.96655s
[20:50:06]	Epoch: 0015/0050	Batch: 0061/0064	Training-Loss: 0.0176648907	Time-Interval: 11.90096s
[20:50:18]	Epoch: 0015/0050	Batch: 0062/0064	Training-Loss: 0.0173247159	Time-Interval: 12.00511s
[20:50:30]	Epoch: 0015/0050	Batch: 0063/0064	Training-Loss: 0.0210063998	Time-Interval: 11.86607s
[20:50:42]	Epoch: 0015/0050	Batch: 0064/0064	Training-Loss: 0.0129876565	Time-Interval: 12.01524s

Hybrid-SIM Training Visualization

[Training Loss] versus [Batch Number] curve (left) and, [Validation Loss] versus [Epoch Number] curve (right).



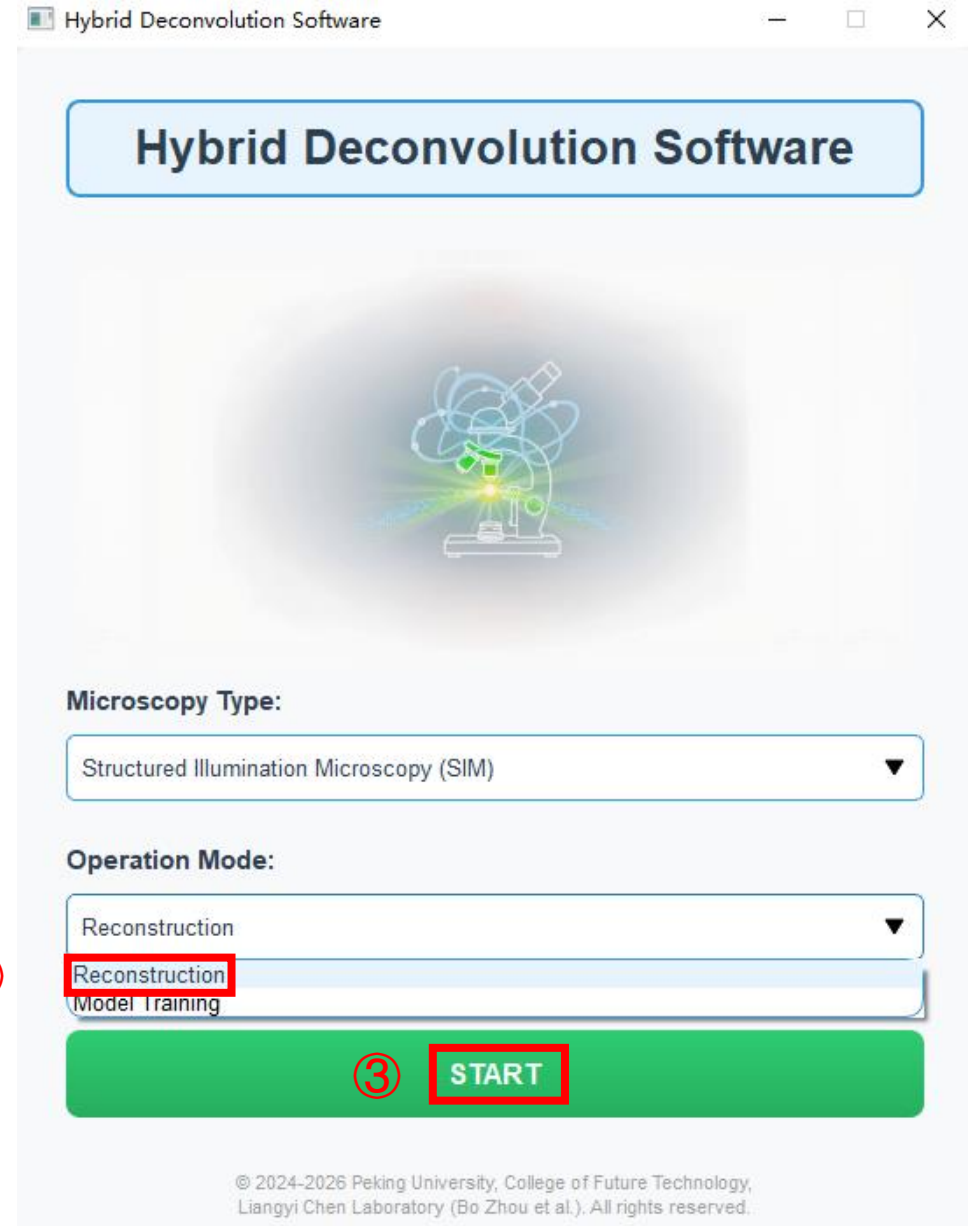
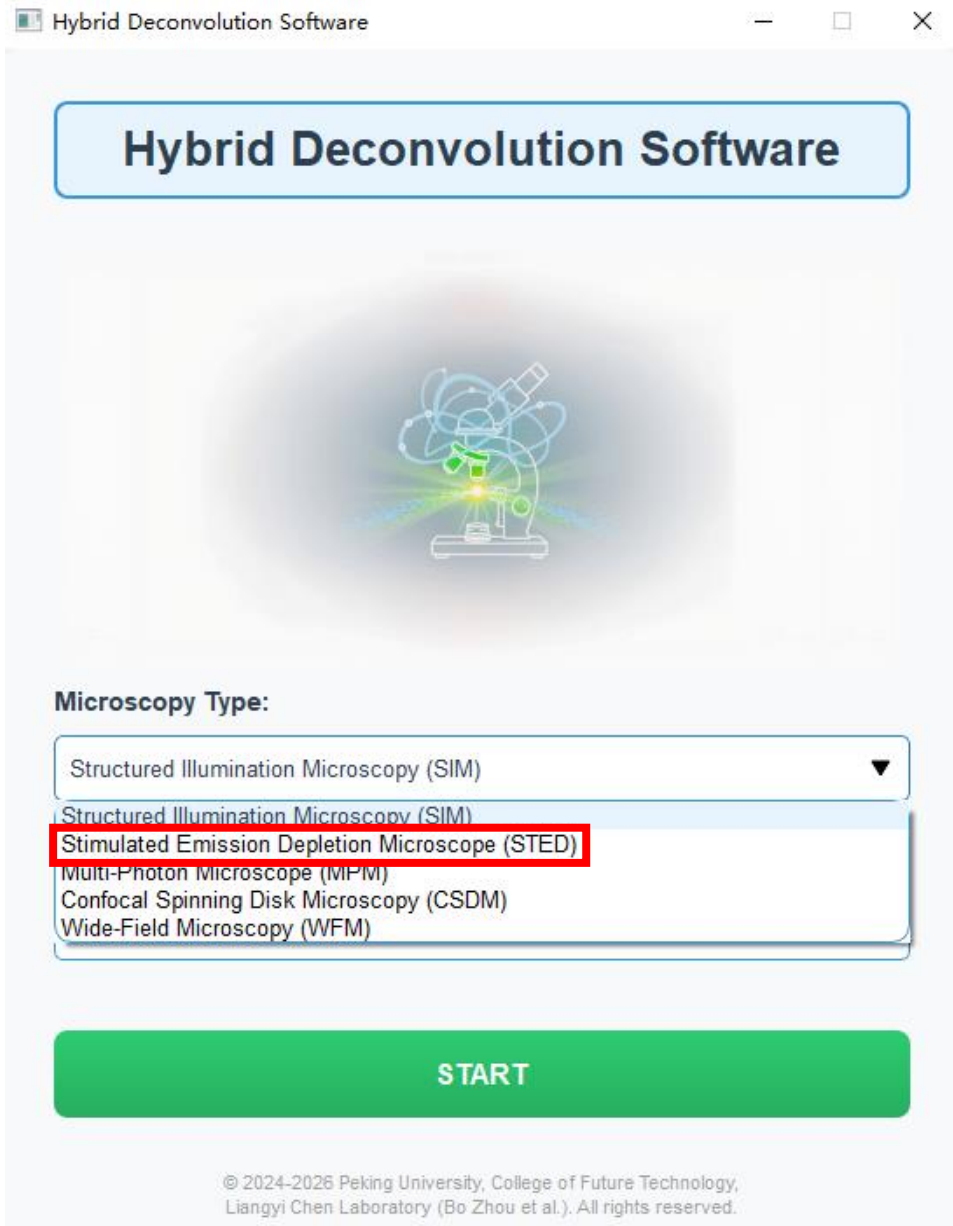
Sample input, output, and ground truth images of the TDV-DNN from the validation dataset are updated at the end of each training epoch.

Please adjust and reset the image display range, zoom view, and frame index here.

Please click the button to reveal the trained model (PTH file) in the Explorer.

Hybrid-STED Reconstruction

Start Hybrid-STED Reconstruction GUI



Hybrid-STED Reconstruction GUI Layout

① Select parameters

The screenshot displays the 'Hybrid-STED Reconstruction' software interface. The window is divided into two main panels: 'Hybrid-STED Reconstruction Parameters' on the left and 'Reconstruction Results' on the right.

Hybrid-STED Reconstruction Parameters Panel:

- At the top, there are three tabs: 'Hessian Denoise', 'TDV Denoise', and 'Sparse Deconvolution'. The 'Sparse Deconvolution' tab is currently selected.
- Below the tabs, the 'Sparse Deconvolution Parameters' section contains several input fields with 'Reset' buttons next to them:
 - Rolling Ball Radius (pixels): 30.00
 - Emission Wavelength (nm): 608.00
 - Raw Pixel Size (nm): 15.00
 - Sparse NA: 6.00
 - Sparse Offset: 0.05
 - Sparse Iteration Number: 20
- A red text box highlights the 'Rolling Ball Radius' parameter, stating: 'Rolling Ball Radius: Radius of curvature of the paraboloid during rolling ball background filtering.'
- At the bottom, there is a 'Device' dropdown menu set to 'cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)' and a 'Detected: 3 device(s)' label. Below this are three buttons: 'Default Parameters', 'Load Parameters', and 'Export Parameters'.
- A large green 'Run Reconstruction' button is prominently displayed.
- Below the green button is a grey 'Stop Reconstruction' button.
- At the very bottom, a green progress bar shows '100%' completion, with the text 'Reconstruction completed!' below it.

Reconstruction Results Panel:

- This panel displays four grayscale microscopy images in a 2x2 grid:
 - Top-left: 'Hessian-STED Reconstruction'
 - Top-right: 'Hessian+TDV-STED Reconstruction'
 - Bottom-left: 'Hybrid-STED Background'
 - Bottom-right: 'Hybrid-STED Reconstruction'
- Below the images, there is a 'Display Range' slider and a 'Frame' slider (set to 1/10).
- Buttons for 'Reset Range', 'Reset View', and 'Export Results' are located at the bottom right of the results panel.

④ Visualize and export results

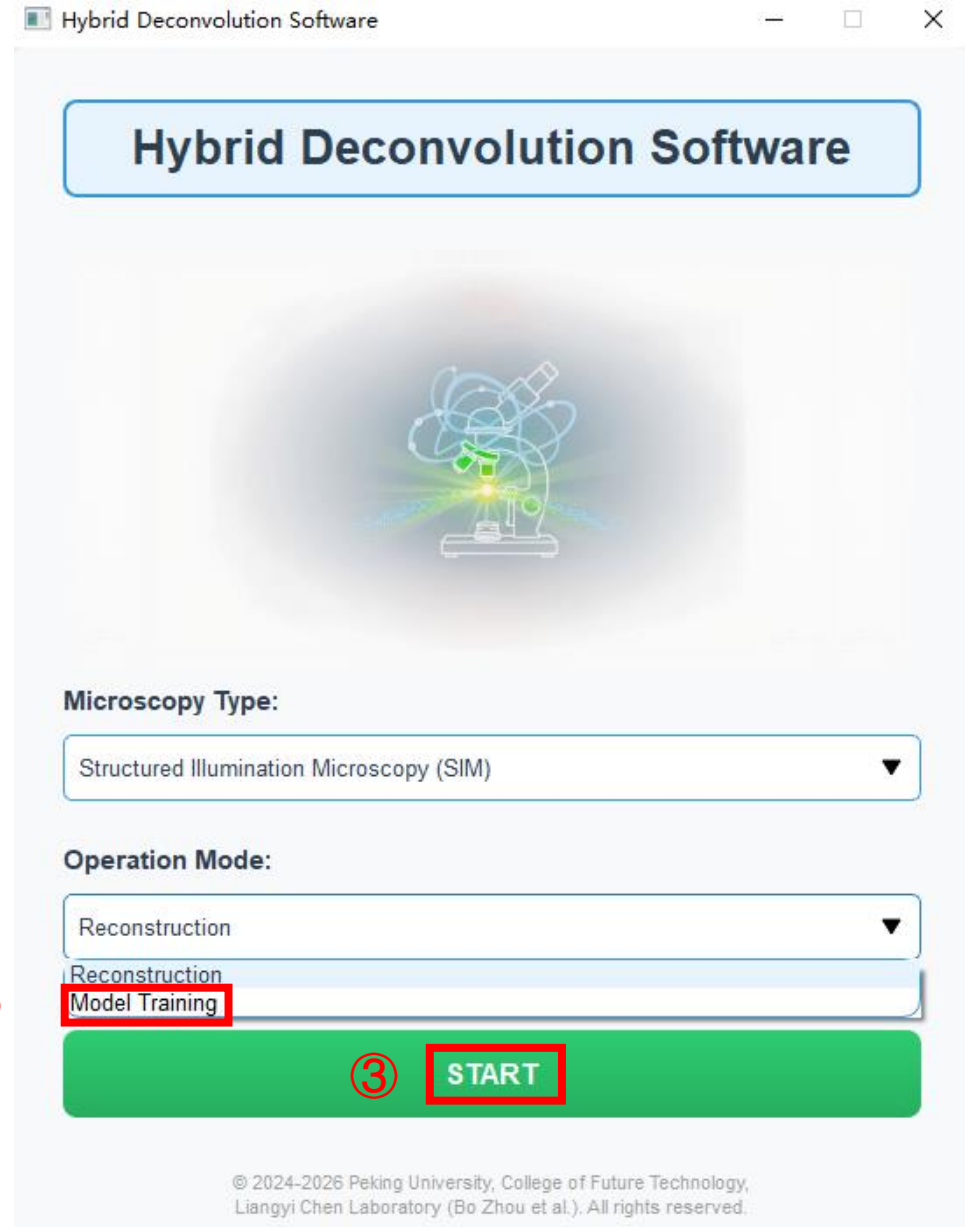
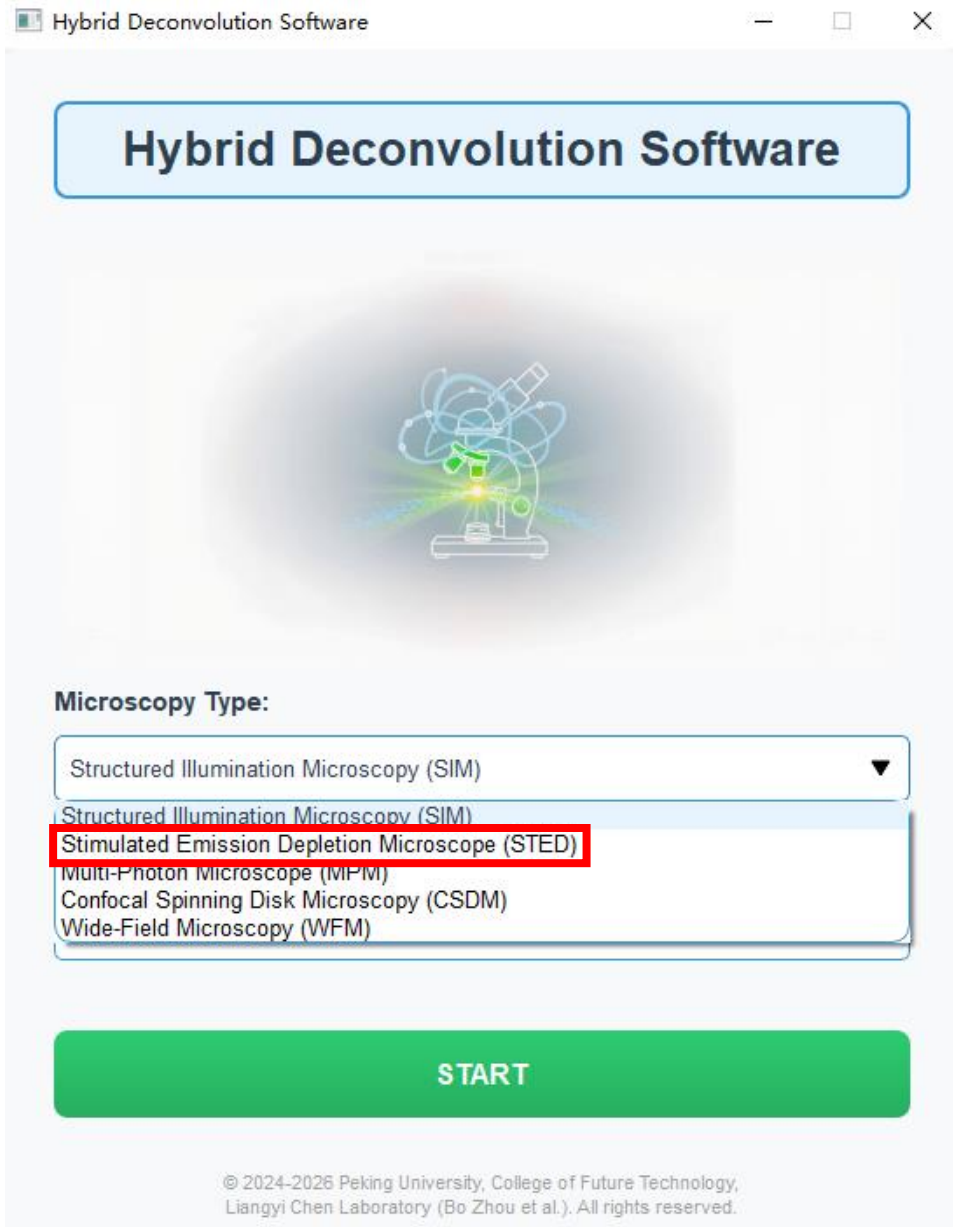
② Run

③ Wait for progress

Tip: The remaining parameters have been covered in the Hybrid-SIM section.

Hybrid-STED Model Training

Start Hybrid-STED Model Training GUI



Hybrid-STED Training GUI layout

① Select parameters

The screenshot displays the Hybrid-STED Training GUI, which is divided into two main panels. The left panel, titled 'Hybrid-STED Training Parameters', contains two tabs: 'Dataset Generation' and 'TDV-DNN Training'. The 'Dataset Generation' tab is active, showing various parameters for dataset creation, including 'Raw Data Folder Path', 'Emission NA', 'Emission Wavelength (nm)', 'Raw Pixel Size (nm)', 'Dataset Z Average Number (frame)', 'Dataset Z Skip Number (frame)', 'Dataset XY Block Size (pixel)', 'Dataset XY Block Interval (pixel)', 'Dataset XY Poisson Noise Level', 'Dataset XY Gaussian Noise Level', and 'Dataset XY Minimal Heterogeneity'. Each parameter has a corresponding input field and a 'Reset' button. Below these parameters are sections for 'Device' (set to 'cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)') and 'Operation' (set to 'Dataset Generation & TDV-DNN Training'). At the bottom of this panel are buttons for 'Default Parameters', 'Load Parameters', 'Export Parameters', a large blue 'Start Training' button, and a grey 'Stop Training' button. A progress bar at the very bottom shows 4% completion. The right panel, titled 'Hybrid-STED Training Visualization', contains two sub-sections. The top section, 'Training Curves', displays two line graphs: 'Training Loss (x0.1)' vs 'Batch Number' (showing a sharp drop from 1.25 to near 0.1) and 'Validation Loss (x0.1)' vs 'Epoch Number' (showing a steady decrease from 0.05 to 0.04). The bottom section, 'Training Results', shows three side-by-side images: 'TDV-DNN Input', 'TDV-DNN Output', and 'Ground Truth'. Below these images are sliders for 'Display Range' and 'Frame' (set to 45/100), along with buttons for 'Reset Range', 'Reset View', and 'Check Model'. At the bottom of the right panel is a 'Training Log' section displaying a list of training events, including epoch and batch numbers, training loss, and time intervals.

Hybrid-STED Training

Hybrid-STED Training Parameters

Dataset Generation TDV-DNN Training

Dataset Generation Parameters

Raw Data Folder Path /Demo_training_STED Browse Reset

Emission NA 5.00 Reset

Emission Wavelength (nm) 608.00 Reset

Raw Pixel Size (nm) 15.00 Reset

Dataset Z Average Number (frame) 1 Reset

Dataset Z Skip Number (frame) 1 Reset

Dataset XY Block Size (pixel) 256 Reset

Dataset XY Block Interval (pixel) 64 Reset

Dataset XY Poisson Noise Level 300.00 Reset

Dataset XY Gaussian Noise Level 300.00 Reset

Dataset XY Minimal Heterogeneity 0.60 Reset

Device: cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB) Detected: 3 device(s)

Operation: Dataset Generation & TDV-DNN Training

Default Parameters Load Parameters Export Parameters

Start Training

Stop Training

4%

Hybrid-STED Training Visualization

Training Curves

Training Loss (x0.1) Batch Number

Validation Loss (x0.1) Epoch Number

Training Results

TDV-DNN Input TDV-DNN Output Ground Truth

Display Range: Reset Range Reset View Check Model

Frame: 45/100

Training Log

[08:33:54] Epoch: 0003/0050 | Batch: 0006/0066 | Training-Loss: 0.0039877440 | Time-Interval: 11.18865s
[08:34:08] Epoch: 0003/0050 | Batch: 0007/0066 | Training-Loss: 0.0040806774 | Time-Interval: 13.80515s
[08:34:21] Epoch: 0003/0050 | Batch: 0008/0066 | Training-Loss: 0.0039975550 | Time-Interval: 13.17471s
[08:34:32] Epoch: 0003/0050 | Batch: 0009/0066 | Training-Loss: 0.0039483020 | Time-Interval: 11.07616s
[08:34:43] Epoch: 0003/0050 | Batch: 0010/0066 | Training-Loss: 0.0036430396 | Time-Interval: 11.17583s
[08:34:54] Epoch: 0003/0050 | Batch: 0011/0066 | Training-Loss: 0.0038224459 | Time-Interval: 10.99109s
[08:35:05] Epoch: 0003/0050 | Batch: 0012/0066 | Training-Loss: 0.0049869474 | Time-Interval: 10.86387s
[08:35:07] Stop TDV-DNN training ...
[08:35:15] TDV-DNN training stopped!
[08:35:15] Ready!

④ Visualize training process and export TDV-DNN model

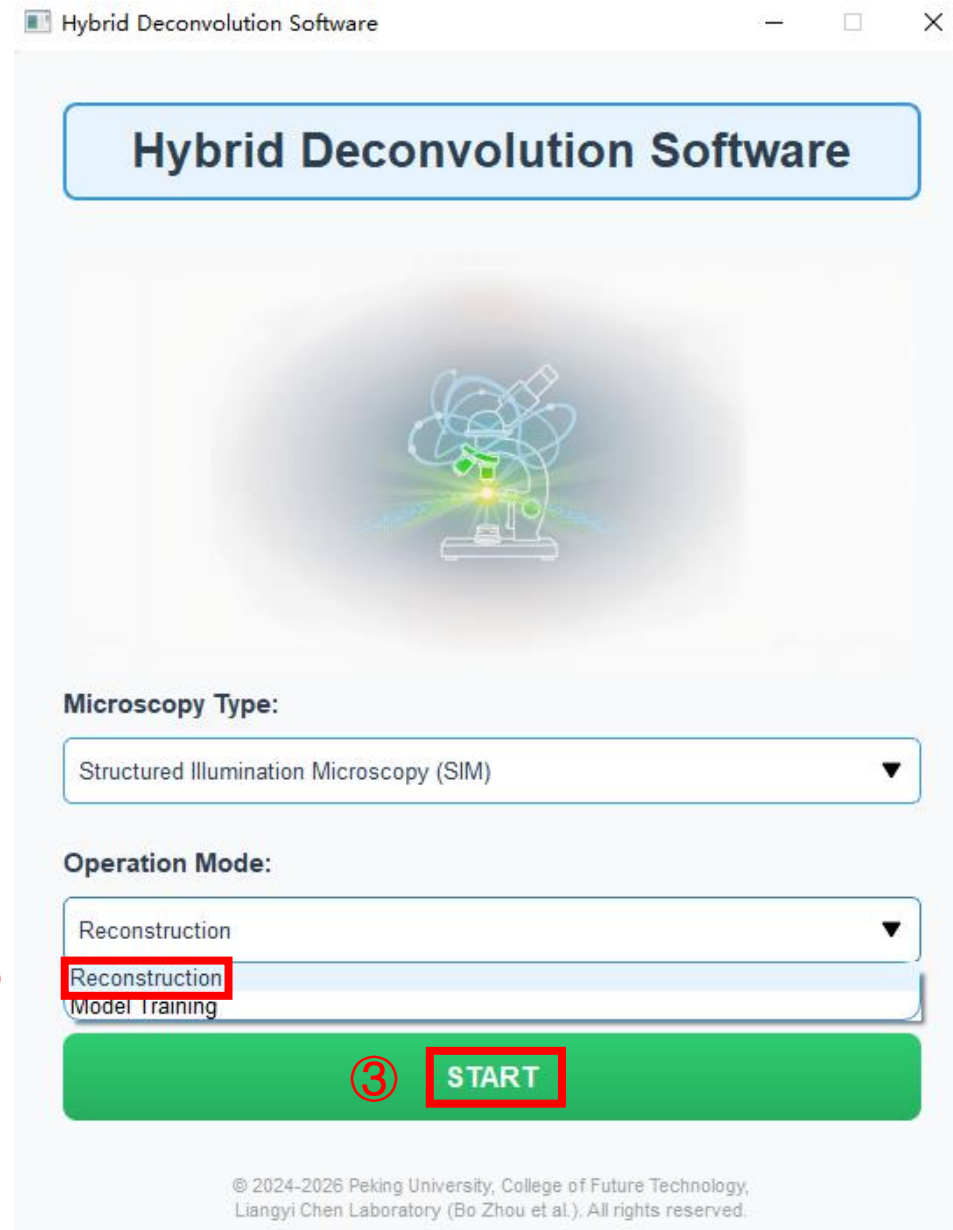
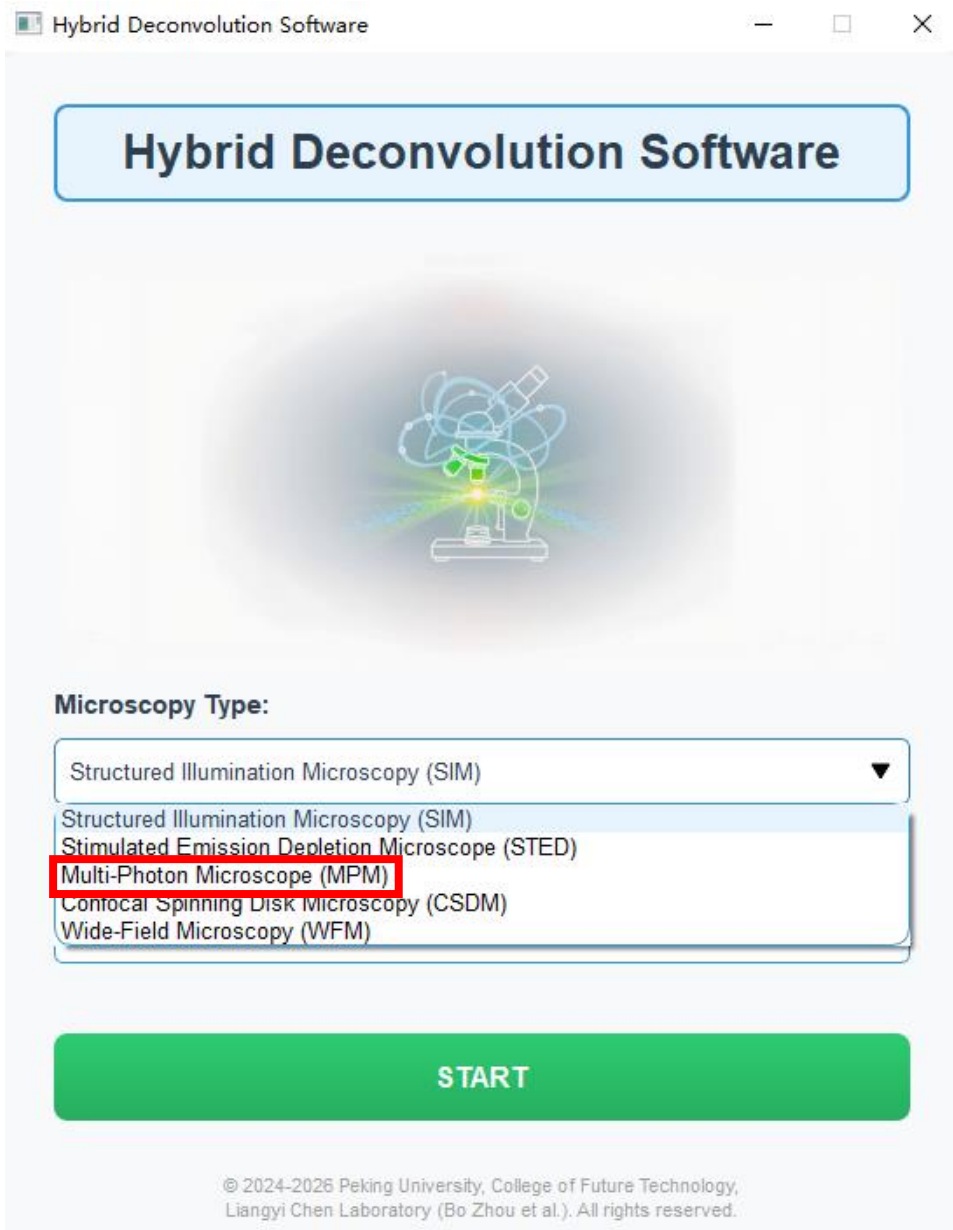
② Run

③ Wait for progress

Tip: The parameters have been covered in the Hybrid-SIM section.

Hybrid-MPM Reconstruction

Start Hybrid-MPM Reconstruction GUI



Hybrid-MPM Reconstruction GUI Layout

① Select parameters

The screenshot displays the 'Hybrid-MPM Reconstruction' application window. The interface is divided into two main panels: 'Hybrid-MPM Reconstruction Parameters' on the left and 'Reconstruction Results' on the right.

Hybrid-MPM Reconstruction Parameters Panel:

- At the top, there are three tabs: 'Hessian Denoise', 'TDV Denoise', and 'Sparse Deconvolution' (which is currently selected).
- Below the tabs, the 'Sparse Deconvolution Parameters' section contains several input fields with 'Reset' buttons:
 - Rolling Ball Radius (pixels): 10.00
 - Emission Wavelength (nm): 525.00
 - Raw Pixel Size (nm): 200.00
 - Sparse NA: 0.30
 - Sparse Offset: 0.02
 - Sparse Iteration Number: 25
- A red text box highlights the 'Rolling Ball Radius' parameter, stating: 'Rolling Ball Radius: Radius of curvature of the paraboloid during rolling ball background filtering.'
- At the bottom, there is a 'Device' dropdown menu set to 'cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)' and a 'Detected: 3 device(s)' label.
- Below the device selection are three buttons: 'Default Parameters', 'Load Parameters', and 'Export Parameters'.
- A large green button labeled 'Run Reconstruction' is prominently displayed.
- Below it is a grey button labeled 'Stop Reconstruction'.
- At the very bottom, a green progress bar shows '100%' completion, with the text 'Reconstruction completed!' below it.

Reconstruction Results Panel:

- This panel displays four grayscale images in a 2x2 grid:
 - Top-left: 'Hessian-MPM Reconstruction' showing a noisy, low-contrast image of a cellular structure.
 - Top-right: 'Hessian+TDV-MPM Reconstruction' showing a clearer, higher-contrast image of the same structure.
 - Bottom-left: 'Hybrid-MPM Background' showing a blurred, low-contrast background image.
 - Bottom-right: 'Hybrid-MPM Reconstruction' showing the final reconstructed image, which is sharp and clear.
- Below the images, there is a 'Display Range' slider and a 'Frame' slider (set to 62/100).
- At the bottom right of this panel are buttons for 'Reset Range', 'Reset View', and 'Export Results'.

④ Visualize and export results

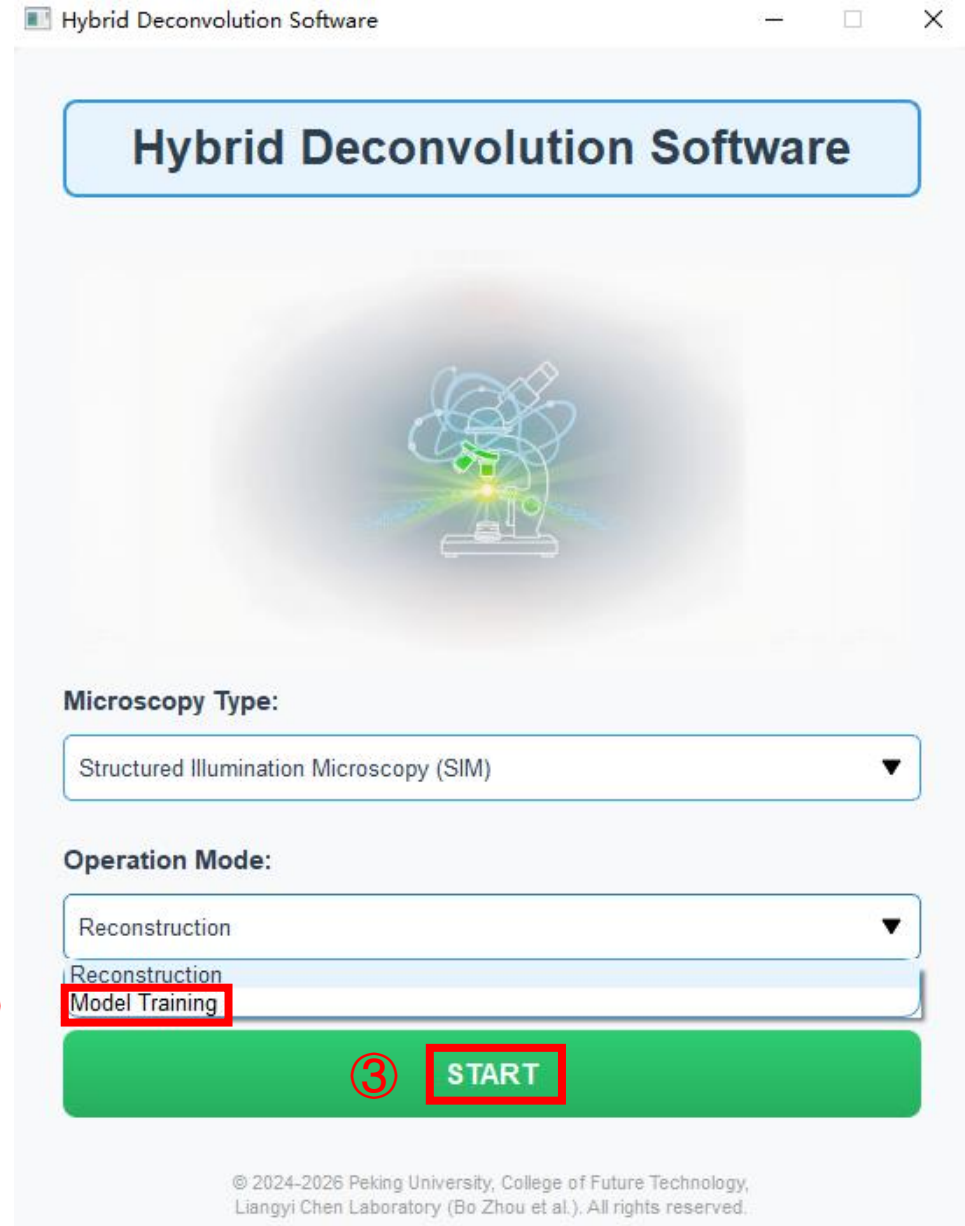
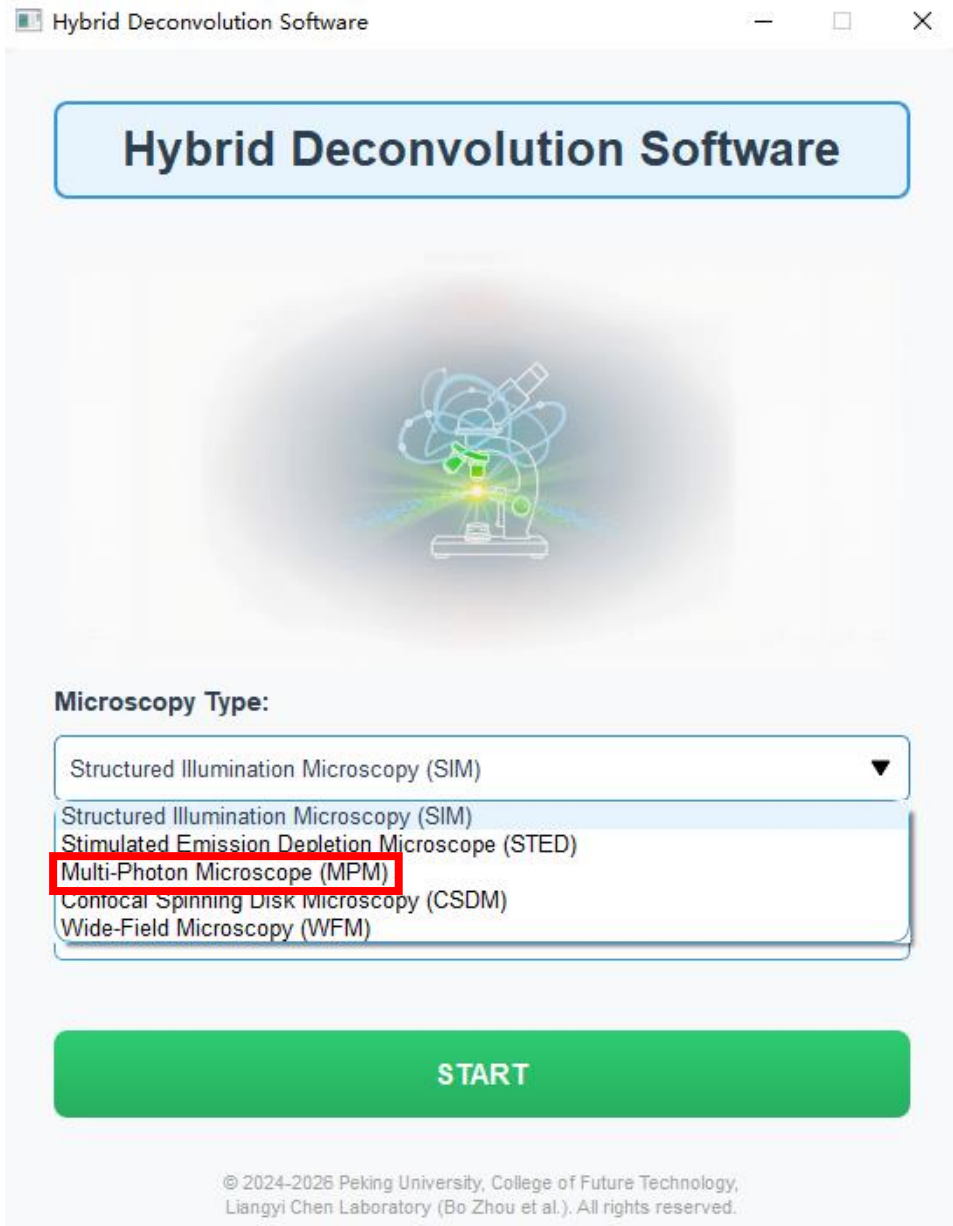
② Run

③ Wait for progress

Tip: The remaining parameters have been covered in the Hybrid-SIM section.

Hybrid-MPM Model Training

Start Hybrid-MPM Model Training GUI



Hybrid-MPM Training GUI layout

① Select parameters

The screenshot displays the Hybrid-MPM Training GUI, which is divided into two main panels: 'Hybrid-MPM Training Parameters' on the left and 'Hybrid-MPM Training Visualization' on the right.

Hybrid-MPM Training Parameters Panel:

- Dataset Generation Parameters:** Includes fields for Raw Data Folder Path (./Demo_training_MPM), Emission NA (0.60), Emission Wavelength (nm) (525.00), Raw Pixel Size (nm) (200.00), Dataset Z Average Number (frame) (20), Dataset Z Skip Number (frame) (10), Dataset XY Block Size (pixel) (256), Dataset XY Block Interval (pixel) (64), Dataset XY Poisson Noise Level (300.00), Dataset XY Gaussian Noise Level (300.00), and Dataset XY Minimal Heterogeneity (0.60). Each field has a 'Reset' button.
- Device and Operation:** A dropdown menu for 'Device' (cuda:1 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)) and a dropdown for 'Operation' (TDV-DNN Training).
- Buttons:** 'Default Parameters', 'Load Parameters', 'Export Parameters', 'Start Training' (highlighted with a red box), and 'Stop Training'.
- Progress Bar:** A progress bar at the bottom shows 10% completion.

Hybrid-MPM Training Visualization Panel:

- Training Curves:** Two line graphs. The left graph shows 'Training Loss (x0.1)' vs 'Batch Number' (0 to 400), with a blue line decreasing from 1.25 to near 0. The right graph shows 'Validation Loss (x0.1)' vs 'Epoch Number' (0 to 5), with a green line decreasing from 0.04 to near 0.03.
- Training Results:** Three side-by-side images labeled 'TDV-DNN Input', 'TDV-DNN Output', and 'Ground Truth'. Below them are sliders for 'Display Range' and 'Frame' (36/100), and buttons for 'Reset Range', 'Reset View', and 'Check Model'.
- Training Log:** A scrollable text area showing training progress, including epoch, batch, training loss, and time interval.

④ Visualize training process and export TDV-DNN model

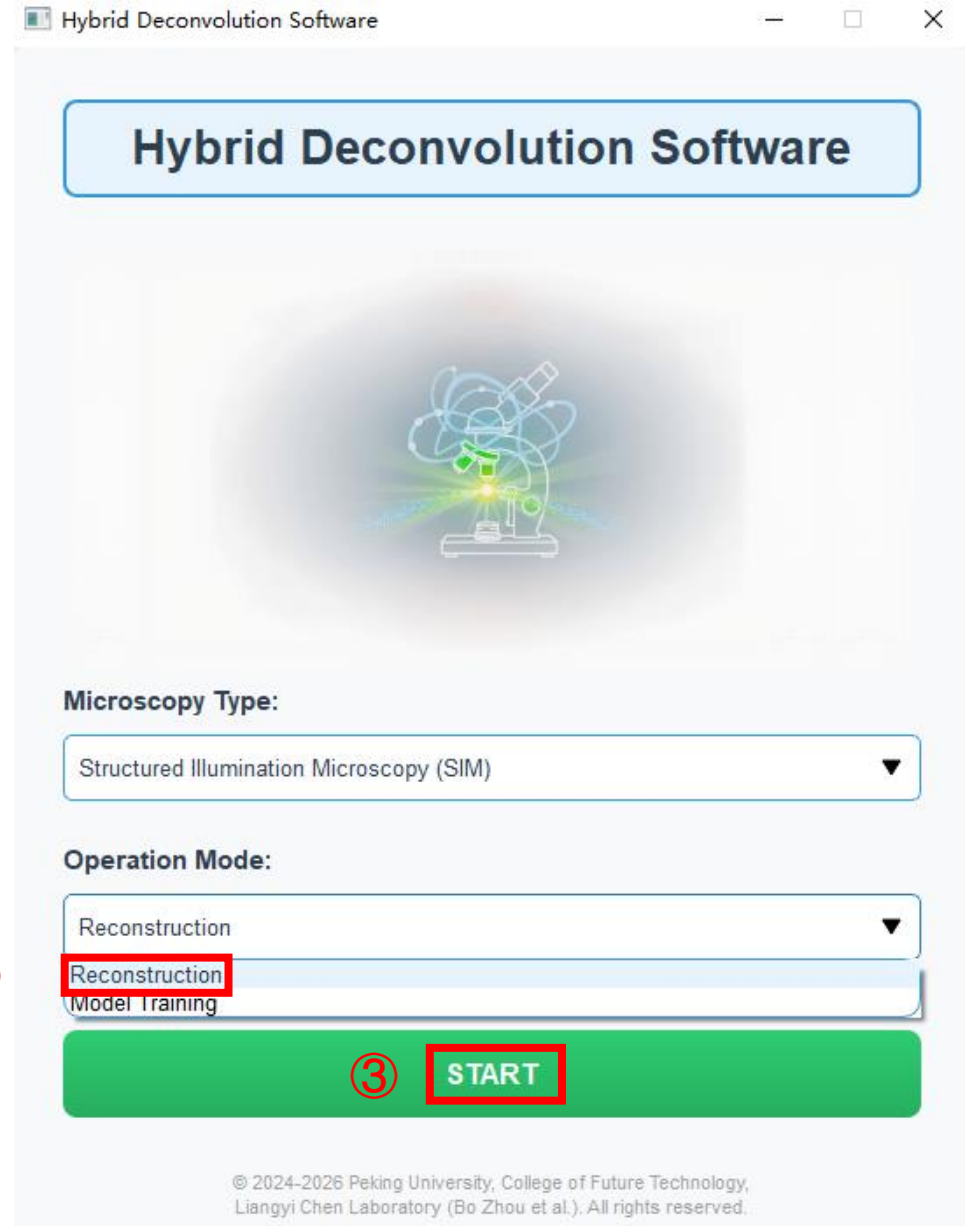
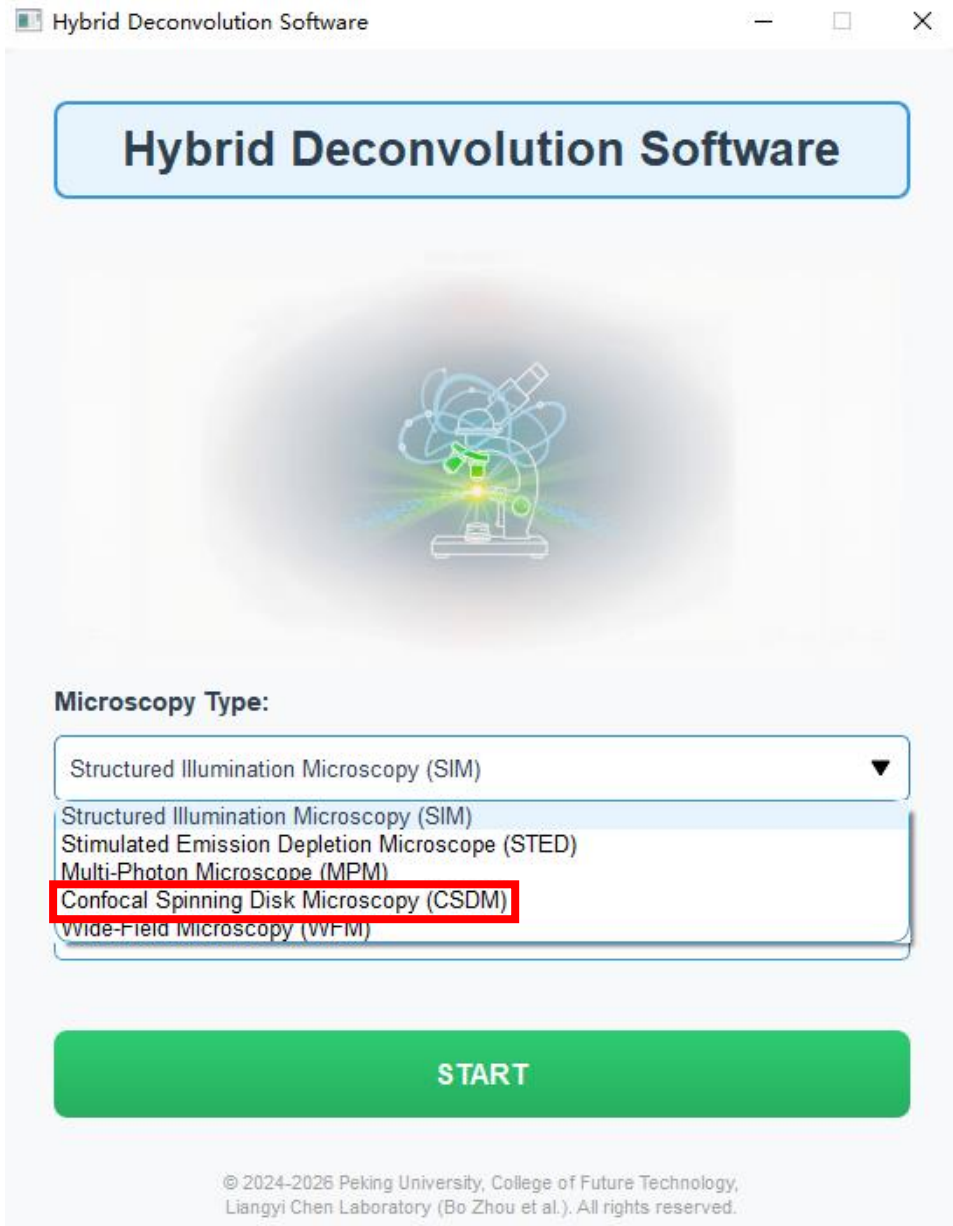
② Run

③ Wait for progress

Tip: The parameters have been covered in the Hybrid-SIM section.

Hybrid-CSDM Reconstruction

Start Hybrid-CSDM Reconstruction GUI



Hybrid-CSDM Reconstruction GUI Layout

① Select parameters

The screenshot displays the 'Hybrid-CSDM Reconstruction' GUI. The interface is divided into two main panels: 'Hybrid-CSDM Reconstruction Parameters' on the left and 'Reconstruction Results' on the right.

Hybrid-CSDM Reconstruction Parameters Panel:

- At the top, there are three tabs: 'Hessian Denoise', 'TDV Denoise', and 'Sparse Deconvolution' (which is currently selected).
- Below the tabs, the 'Sparse Deconvolution Parameters' section contains several input fields with 'Reset' buttons:
 - Rolling Ball Radius (pixels): 100.00
 - Emission Wavelength (nm): 525.00
 - Raw Pixel Size (nm): 65.00
 - Sparse NA: 1.20
 - Sparse Offset: 0.05
 - Sparse Iteration Number: 20
- Below these fields, a text box explains the 'Rolling Ball Radius':

Rolling Ball Radius:
Radius of curvature of the paraboloid during rolling ball background filtering.
- At the bottom, there is a 'Device' dropdown menu showing 'cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)' and a 'Detected: 3 device(s)' label. Below this are three buttons: 'Default Parameters', 'Load Parameters', and 'Export Parameters'.
- Below the parameters panel, there is a large green button labeled 'Run Reconstruction' and a grey button labeled 'Stop Reconstruction'.
- At the bottom of the parameters panel, there is a green progress bar showing '100%' and the text 'Reconstruction completed!'.

Reconstruction Results Panel:

- This panel displays four grayscale images in a 2x2 grid:
 - Top-left: 'Hessian-CSDM Reconstruction'
 - Top-right: 'Hessian+TDV-CSDM Reconstruction'
 - Bottom-left: 'Hybrid-CSDM Background'
 - Bottom-right: 'Hybrid-CSDM Reconstruction'
- Below the images, there are two sliders for 'Display Range' and a 'Frame' slider. To the right of the sliders are buttons for 'Reset Range', 'Reset View', and 'Export Results'.

④ Visualize and export results

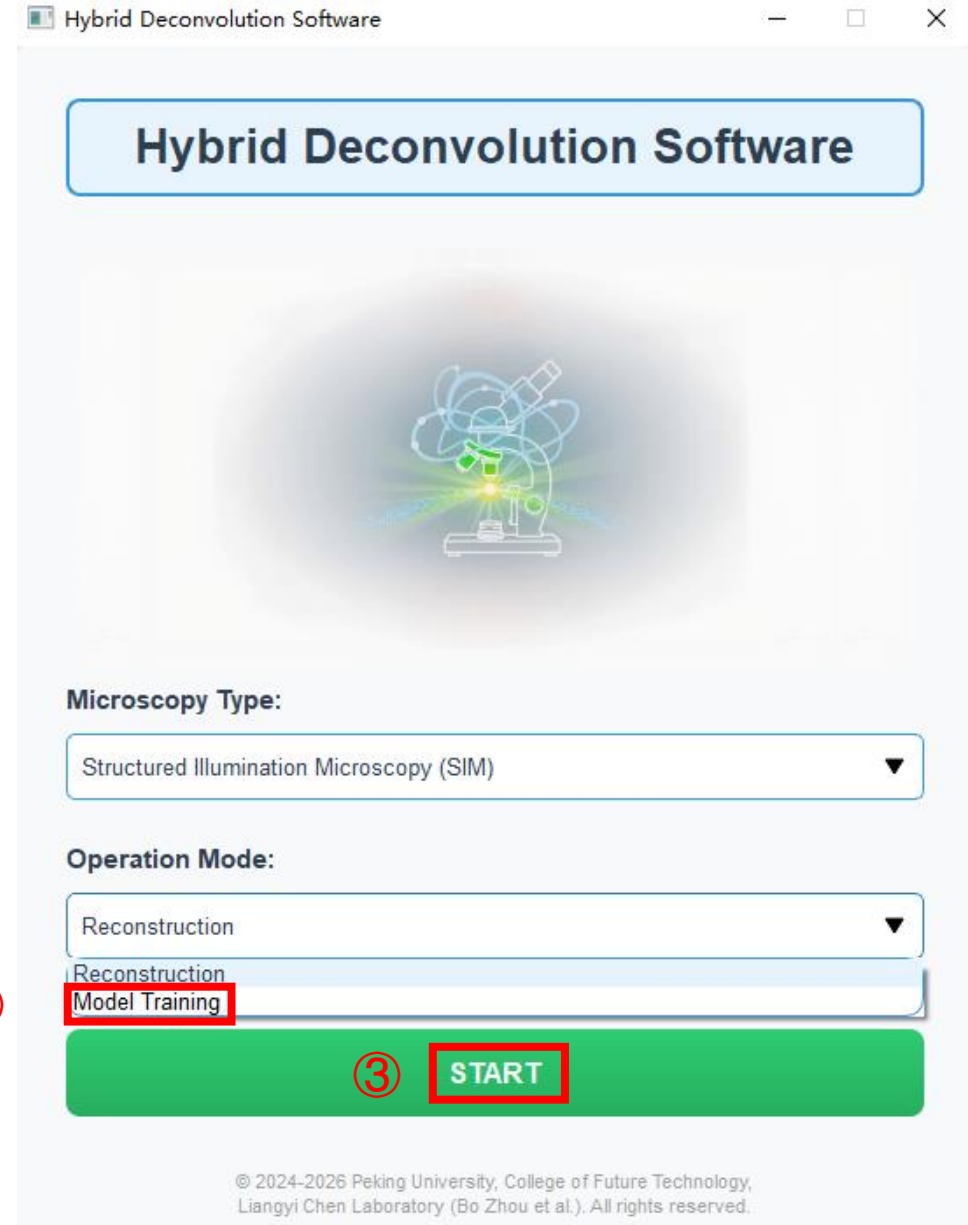
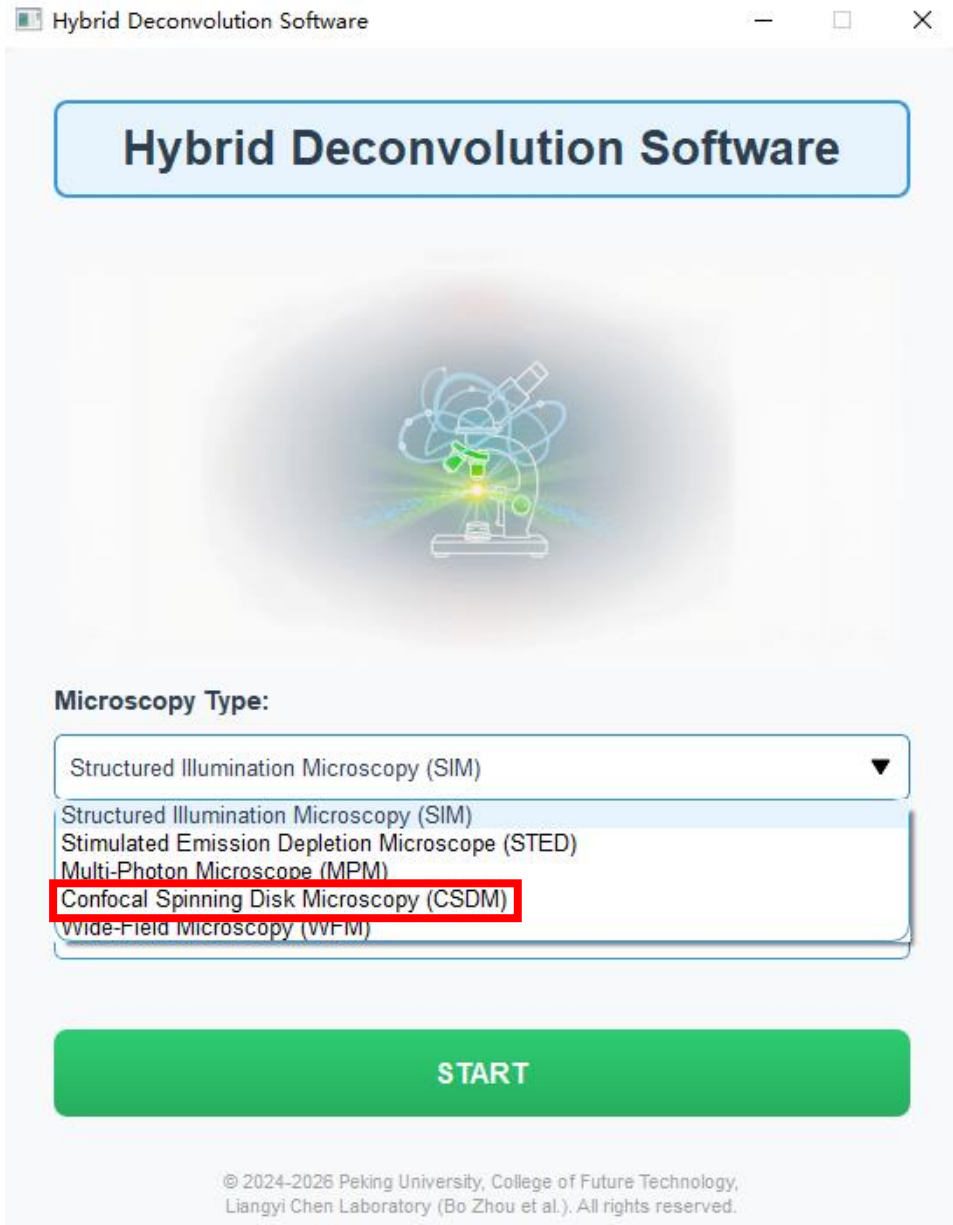
② Run

③ Wait for progress

Tip: The remaining parameters have been covered in the Hybrid-SIM section.

Hybrid-CSDM Model Training

Start Hybrid-CSDM Model Training GUI



Hybrid-CSDM Training GUI layout

① Select parameters

The screenshot displays the Hybrid-CSDM Training GUI, which is divided into two main panels: 'Hybrid-CSDM Training Parameters' on the left and 'Hybrid-CSDM Training Visualization' on the right.

Hybrid-CSDM Training Parameters Panel:

- Dataset Generation Parameters:** Includes fields for Raw Data Folder Path (Demo_training_CSDM), Emission NA (1.30), Emission Wavelength (nm) (525.00), Raw Pixel Size (nm) (65.00), Dataset Z Average Number (frame) (20), Dataset Z Skip Number (frame) (16), Dataset XY Block Size (pixel) (256), Dataset XY Block Interval (pixel) (128), Dataset XY Poisson Noise Level (200.00), Dataset XY Gaussian Noise Level (200.00), and Dataset XY Minimal Heterogeneity (0.60). Each field has a 'Reset' button.
- Device:** A dropdown menu showing 'cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)' with a 'Detected: 3 device(s)' label.
- Operation:** A dropdown menu showing 'Dataset Generation & TDV-DNN Training'.
- Buttons:** 'Default Parameters', 'Load Parameters', 'Export Parameters', 'Start Training' (highlighted with a red box), and 'Stop Training'.
- Progress Bar:** A progress bar at the bottom shows 4% completion.

Hybrid-CSDM Training Visualization Panel:

- Training Curves:** Two line graphs. The left graph shows 'Training Loss (x0.1)' vs 'Batch Number' (0 to 125), with a blue line showing a sharp drop and then leveling off. The right graph shows 'Validation Loss (x0.1)' vs 'Epoch Number' (0.0 to 2.0), with a green line showing a steady decrease.
- Training Results:** Three side-by-side images labeled 'TDV-DNN Input', 'TDV-DNN Output', and 'Ground Truth', showing grayscale microscopy images of a porous material.
- Display Range:** A slider bar with 'Reset Range' and 'Reset View' buttons.
- Frame:** A slider bar showing '52/100'.
- Check Model:** A green button.
- Training Log:** A scrollable text area showing training progress logs, including epoch, batch, training loss, validation loss, and time intervals.

② Run

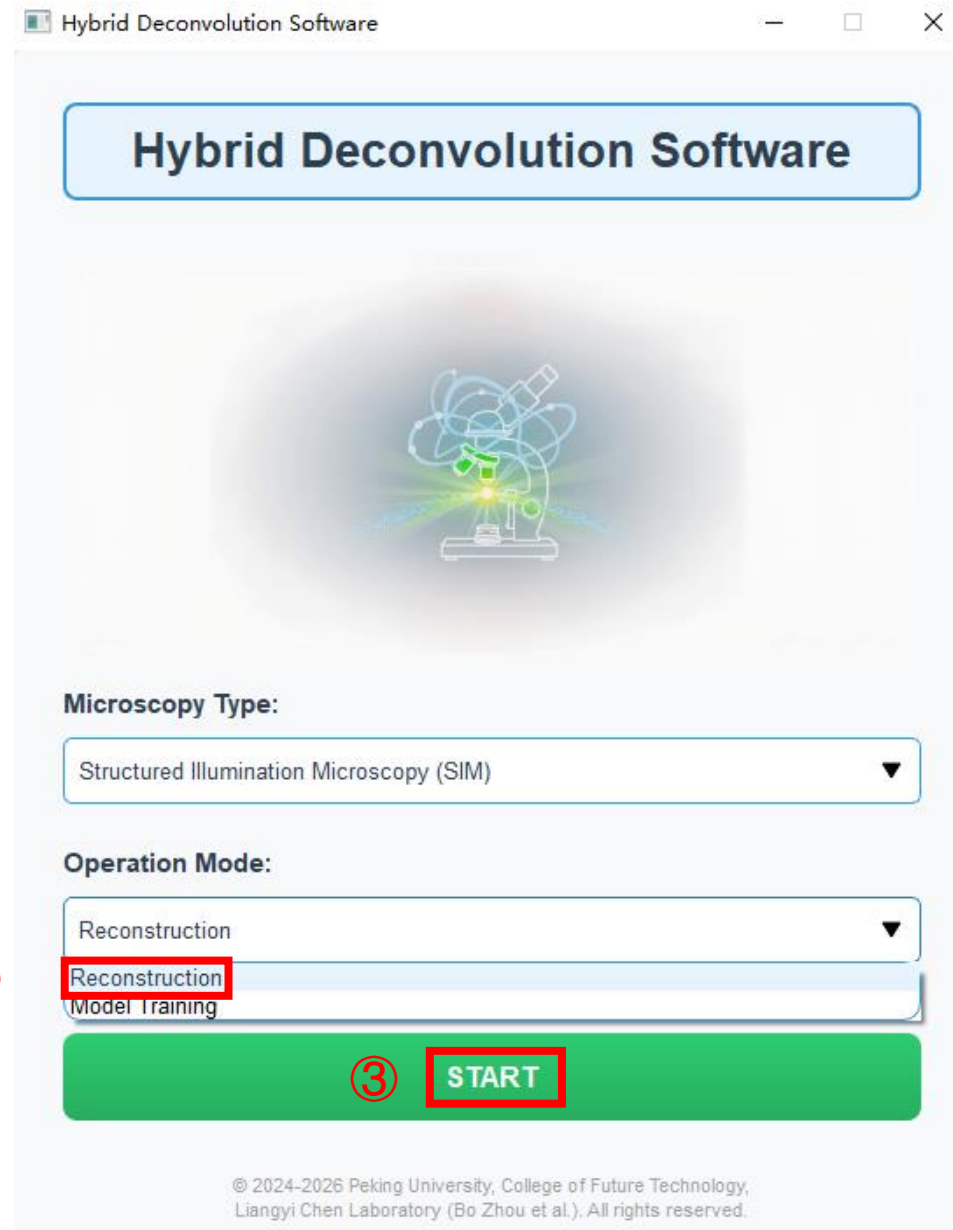
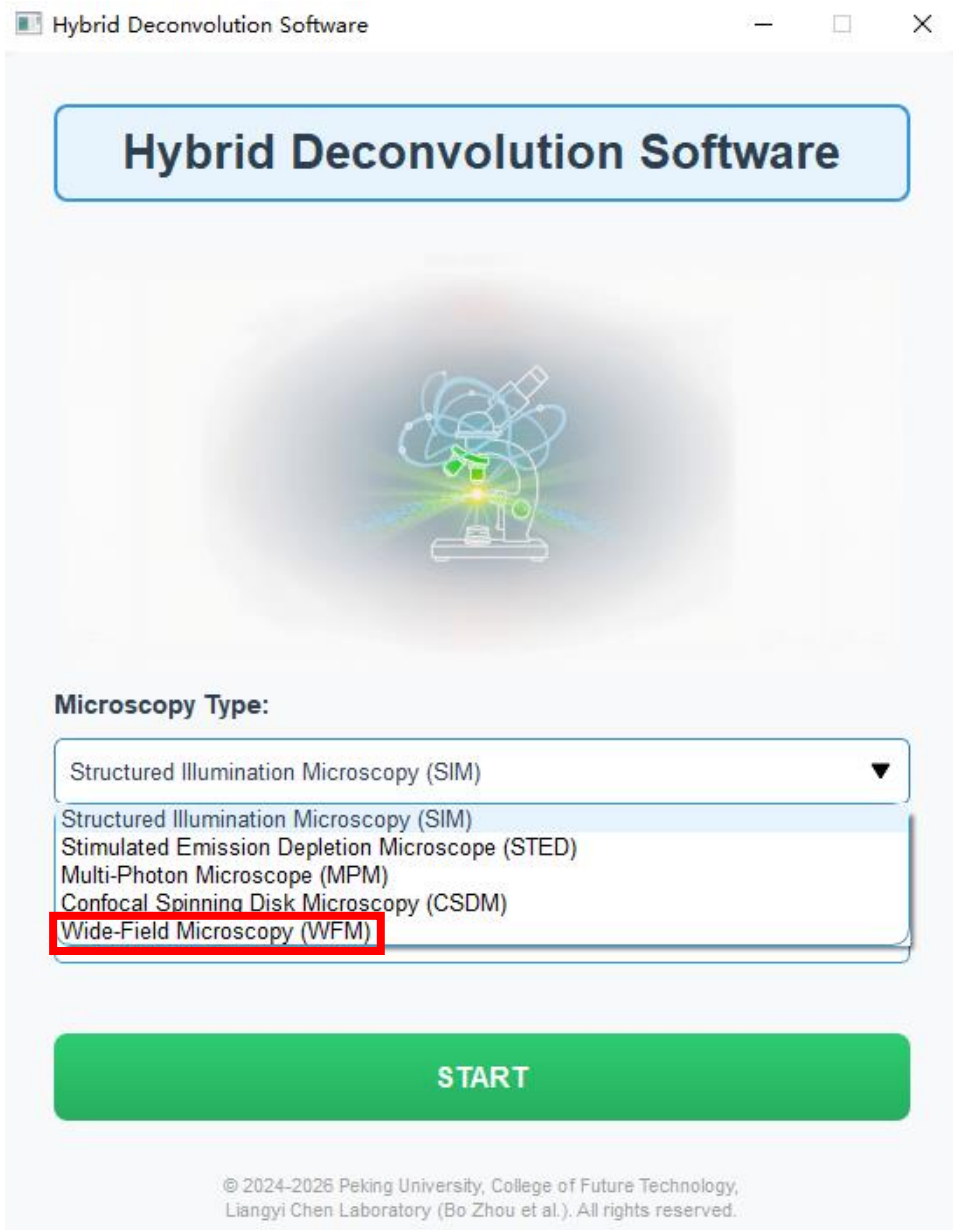
③ Wait for progress

④ Visualize training process and export TDV-DNN model

Tip: The parameters have been covered in the Hybrid-SIM section.

Hybrid-WFM Reconstruction

Start Hybrid-WFM Reconstruction GUI



Hybrid-WFM Reconstruction GUI Layout

① Select parameters

The screenshot displays the 'Hybrid-WFM Reconstruction' application window, which is divided into two main panels: 'Hybrid-WFM Reconstruction Parameters' on the left and 'Reconstruction Results' on the right. The left panel features a tabbed interface with 'BF-WFM' selected, showing input fields for 'WFM Raw Data Path', 'WFM Raw Pixel Size (nm)', 'WFM Emission Wavelength (nm)', and 'WFM Emission NA', each with a 'Reset' button. Below these is a 'Device' dropdown menu and three buttons: 'Default Parameters', 'Load Parameters', and 'Export Parameters'. A large green 'Run Reconstruction' button is prominently displayed, with a red box highlighting it. Below this is a grey 'Stop Reconstruction' button. At the bottom of the left panel, a green progress bar shows '100%' completion, with the text 'Reconstruction completed!' below it. The right panel, titled 'Reconstruction Results', contains four sub-images: 'BF-WFM Reconstruction', 'BF+Hessian-WFM Reconstruction', 'BF+Hessian+TDV-WFM Reconstruction', and 'Hybrid-WFM Reconstruction'. Below these images are sliders for 'Display Range' and 'Frame' (set to 38/100), along with 'Reset Range', 'Reset View', and 'Export Results' buttons. Red arrows and numbers 1 through 4 are overlaid on the image to guide the user through the workflow: 1 points to the parameter selection area, 2 points to the 'Run Reconstruction' button, 3 points to the progress bar, and 4 points to the 'Export Results' button.

Hybrid-WFM Reconstruction

Hybrid-WFM Reconstruction Parameters

BF-WFM Hessian Denoise TDV Denoise Sparse Deconvolution

BF-WFM Parameters

WFM Raw Data Path o_Recon_WFM.tif Browse Reset

WFM Raw Pixel Size (nm) 65.00 Reset

WFM Emission Wavelength (nm) 525.00 Reset

WFM Emission NA 1.40 Reset

Device: cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB) Detected: 3 device(s)

Default Parameters Load Parameters Export Parameters

Run Reconstruction

Stop Reconstruction

100%

Reconstruction completed!

Reconstruction Results

BF-WFM Reconstruction BF+Hessian-WFM Reconstruction

BF+Hessian+TDV-WFM Reconstruction Hybrid-WFM Reconstruction

Display Range: Reset Range Reset View Export Results

Frame: 38/100

④ Visualize and export results

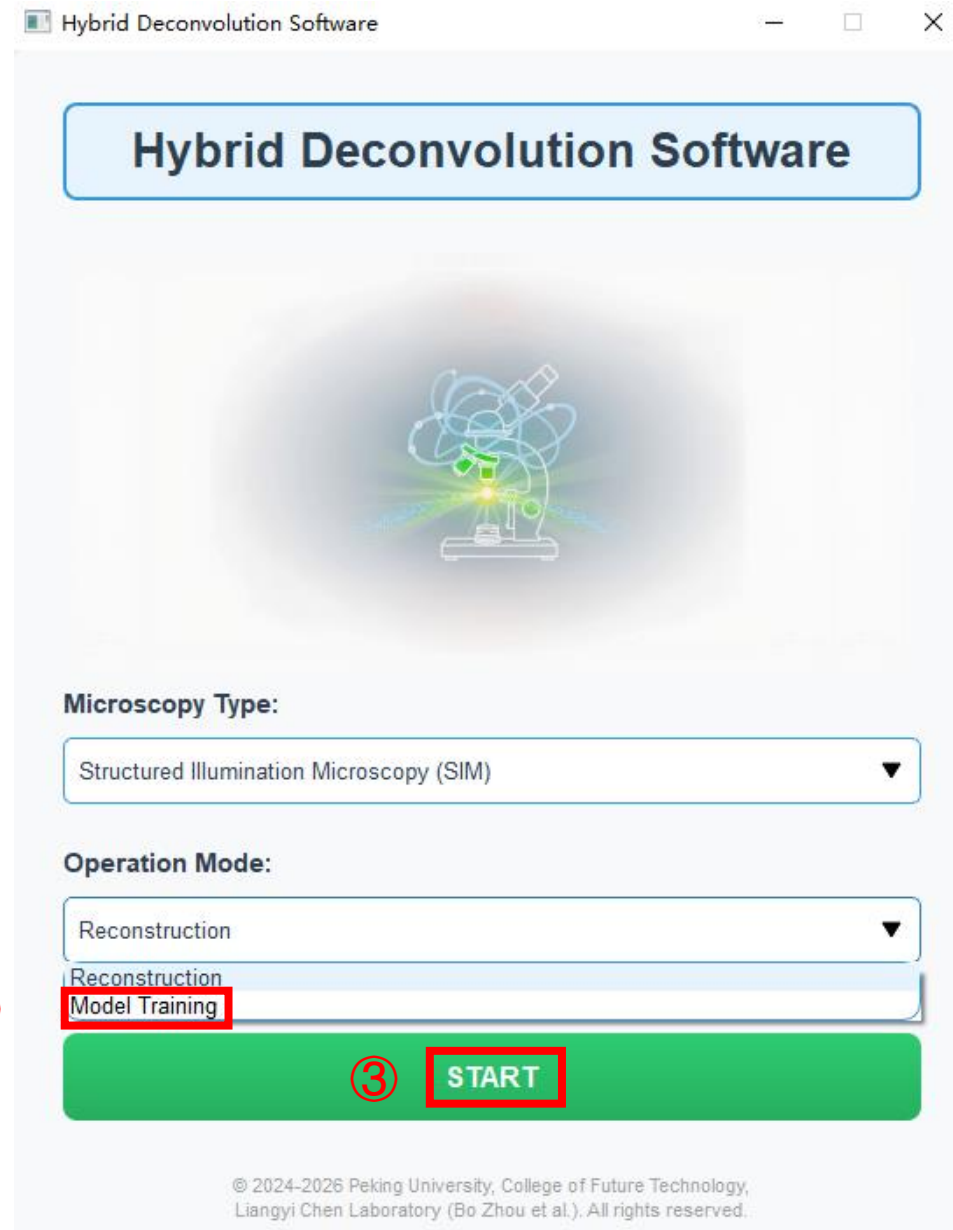
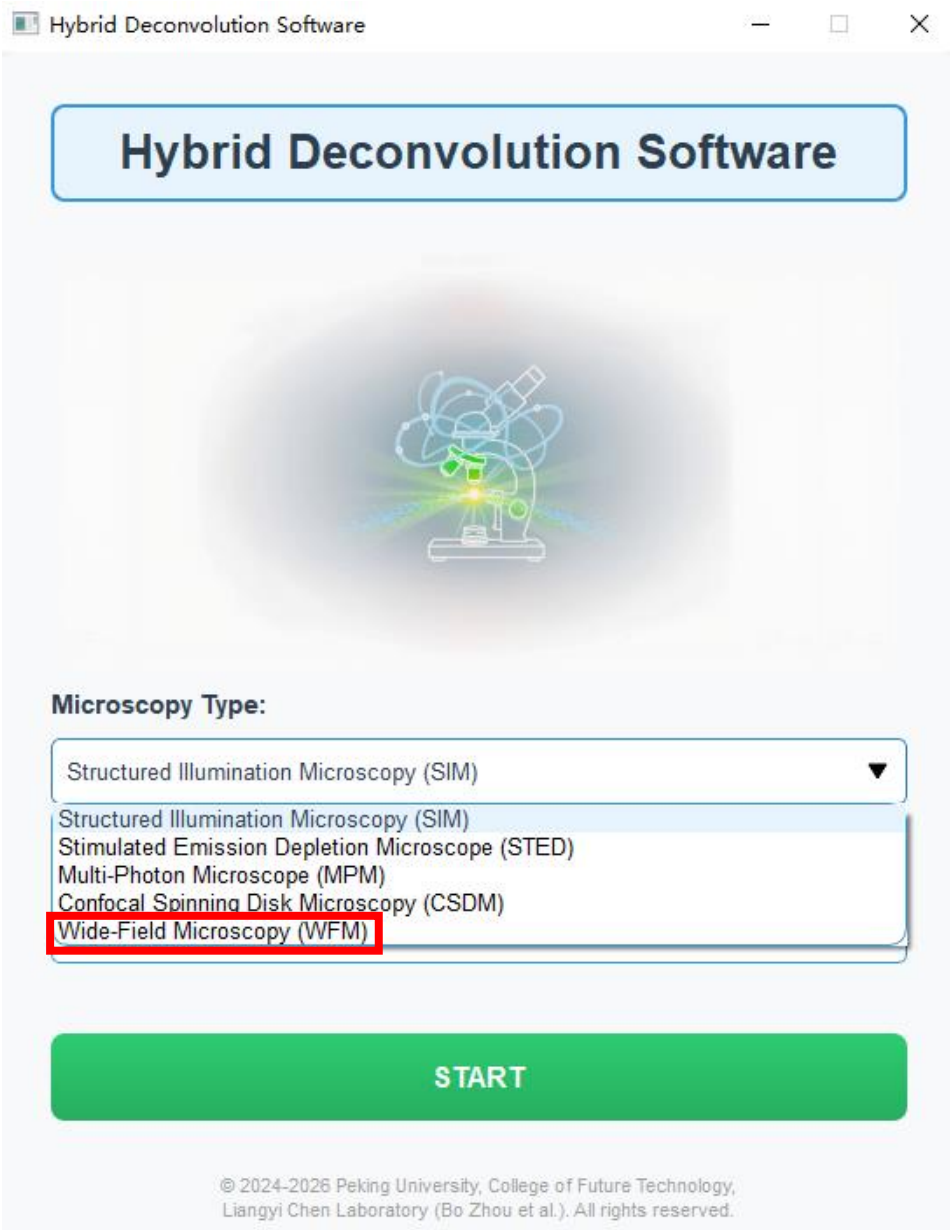
② Run

③ Wait for progress

Tip: The parameters have been covered in the Hybrid-SIM section.

Hybrid-WFM Model Training

Start Hybrid-WFM Model Training GUI



Hybrid-WFM Training GUI layout

① Select parameters

The screenshot displays the 'Hybrid-WFM Training' application window, which is divided into two main panels. The left panel, titled 'Hybrid-WFM Training Parameters', contains two tabs: 'Dataset Generation' and 'TDV-DNN Training'. The 'Dataset Generation' tab is active, showing various parameters for dataset creation, each with a 'Reset' button. At the bottom of this panel are buttons for 'Default Parameters', 'Load Parameters', and 'Export Parameters'. The right panel, titled 'Hybrid-WFM Training Visualization', contains two sub-sections. The top sub-section, 'Training Curves', displays two line graphs: 'Training Loss (x0.1)' vs 'Batch Number' and 'Validation Loss (x0.1)' vs 'Epoch Number'. The bottom sub-section, 'Training Results', shows three side-by-side images: 'TDV-DNN Input', 'TDV-DNN Output', and 'Ground Truth'. Below these images are sliders for 'Display Range' and 'Frame', along with 'Reset Range', 'Reset View', and 'Check Model' buttons. At the bottom of the right panel is a 'Training Log' section showing a list of training events with timestamps, epoch/batch numbers, and loss values. A progress bar at the bottom of the left panel shows the training progress at 10%.

Hybrid-WFM Training Parameters

Dataset Generation | TDV-DNN Training

Dataset Generation Parameters

WFM Raw Data Folder Path: ./Demo_training_WFM [Browse] [Reset]

Emission NA: 1.40 [Reset]

Emission Wavelength (nm): 525.00 [Reset]

Raw Pixel Size (nm): 65.00 [Reset]

Dataset Z Average Number (frame): 20 [Reset]

Dataset Z Skip Number (frame): 5 [Reset]

Dataset XY Block Size (pixel): 256 [Reset]

Dataset XY Block Interval (pixel): 48 [Reset]

Dataset XY Poisson Noise Level: 80.00 [Reset]

Dataset XY Gaussian Noise Level: 80.00 [Reset]

Dataset XY Minimal Heterogeneity: 0.10 [Reset]

Device: cuda:1 (NVIDIA GeForce RTX 3080 Ti, 12.0GB) Detected: 3 device(s)

Operation: Dataset Generation & TDV-DNN Training

[Default Parameters] [Load Parameters] [Export Parameters]

Start Training

Stop Training

10%

Hybrid-WFM Training Visualization

Training Curves

Training Loss (x0.1) vs Batch Number

Validation Loss (x0.1) vs Epoch Number

Training Results

TDV-DNN Input | TDV-DNN Output | Ground Truth

Display Range: [Slider] [Reset Range] [Reset View] [Check Model]

Frame: [Slider] 57/100

Training Log

[11:58:18] Epoch: 0005/0050 | Batch: 0068/0070 | Training-Loss: 0.0003616210 | Time-Interval: 11.73386s
[11:58:29] Epoch: 0005/0050 | Batch: 0069/0070 | Training-Loss: 0.0004052841 | Time-Interval: 11.68205s
[11:58:37] Epoch: 0005/0050 | Batch: 0070/0070 | Training-Loss: 0.0003730508 | Time-Interval: 7.36678s
[11:59:26] Epoch: 0005/0050 | Validation-Loss: 0.0003505269 | Time-Interval: 48.97345s
[11:59:37] Epoch: 0006/0050 | Batch: 0001/0070 | Training-Loss: 0.0003593634 | Time-Interval: 11.73782s
[11:59:49] Epoch: 0006/0050 | Batch: 0002/0070 | Training-Loss: 0.0003981451 | Time-Interval: 11.62355s
[12:00:01] Epoch: 0006/0050 | Batch: 0003/0070 | Training-Loss: 0.0003767489 | Time-Interval: 11.66596s
[12:00:11] Stop TDV-DNN training ...
[12:00:13] TDV-DNN training stopped!
[12:00:13] Ready!

② Run

③ Wait for progress

④ Visualize training process and export TDV-DNN model

Tip: The parameters have been covered in the Hybrid-SIM section.

Thanks!